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A two-stage large-scale multiobjective evolutionary algorithm based on offset direction sampling and dual-layer competition

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ABSTRACT

Large-scale multiobjective optimization problems are characterized by high-dimensional decision spaces and complex search landscapes. These challenges create a dilemma for balancing convergence and diversity in the objective space under limited function evaluations. To address this issue, this paper proposes a two-stage large-scale multiobjective evolutionary algorithm based on offset direction sampling and dual-layer competition. In the rapid convergence stage, a set of high-quality solutions is used to construct search directions in the decision space, followed by sampling along these directions to assist faster convergence of the population. In the precise convergence stage, all individuals are divided into elite and non-elite layers based on non-dominated ranks. Through competition between different layers, two outstanding individuals are identified to further guide the population toward the Pareto-optimal front. Experimental evaluations on benchmark demonstrate significant competitive advantage of the proposed algorithm in addressing large-scale multiobjective optimization challenges involving up to 5000 decision variables.

1. Introduction

Multiobjective optimization problems (MOPs) are prevalent in various real-world applications (Sabir et al., 2024a; Sabir et al., 2024b). Due to the conflicts among objectives, identifying a solution that optimally satisfies all objective functions is generally unattainable. Achieving a balance among these objectives necessitates multiple distinct solutions. To address MOPs, researchers have developed population-based evolutionary algorithms (Coello et al., 2007), which are now widely adopted (AbuAli and Sabir, 2024; Bhat et al., 2024). With the increasing complexity of societal production scenarios and the advent of the big data era, MOPs are becoming more intricate and are trending towards large-scale instances. Large-scale multiobjective optimization problems (LSMOPs), such as complex community detection problems (Liu et al., 2014), capacitated vehicle routing problems (Duan et al., 2021), and deep neural architecture search problems (Lu et al., 2019), can involve thousands of decision variables. As the dimensionality of decision variables grows, the search space experiences exponential expansion. This growth hampers the efficiency of evolutionary algorithms (Tian et al., 2021). To address this challenge, researchers have proposed evolutionary algorithms specifically designed for LSMOPs, named as large-scale multiobjective evolutionary algorithms (LSMOEAs). Currently, LSMOEAs can be classified into these approaches: decision variable grouping, problem transformation, search direction, and novel search strategies. Among these, decision variable grouping methods often adopt a “divide and conquer” method. These algorithms optimize each group alternately, and achieve overall optimization through the coevolution of subpopulations. However, in LSMOPs, interactions among decision variables may vary due to the presence of multiple conflicting objectives. Consequently, some researchers optimize different types of decision variables based on their specific characteristics. LSMOEAs employing problem transformation

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reduce complexity by defining transformation functions, thereby enabling more efficient discovery of high-quality solutions. LSMOEAs focusing on search direction construct efficient search pathways within the decision space, generate sample solutions to facilitate evolution, and enable rapid convergence of the population within limited computational resources. LSMOEAs utilizing novel search strategies does not require partitioning or dimensionality reduction of decision variables. These algorithms employ novel reproduction methods to balance exploration and exploitation within the intricate decision space, thereby expediting the convergence of the population. Overall, research on LSMOEAs has yielded significant results; however, substantial opportunities remain in both convergence and diversity. First, when the number of decision variables is large and complex correlations exist among them, a significant number of function evaluations are required for variable grouping or analysis. Furthermore, the resulting subproblems may still involve hundreds of variables, leading to constrain optimization performance and increased computational costs. Second, problem transformation method restricts the search space of the original decision variables, potentially leading to local optima and thereby missing the global optimum of the original LSMOP. Third, search direction-based LSMOEAs frequently exhibit insufficient diversity in their generated search directions, which constrains their capacity to effectively explore high-dimensional decision spaces. Finally, LSMOEAs employing novel search strategies typically require millions of function evaluations to achieve optimal solutions.

In summary, this paper proposes a two-stage large-scale multiobjective evolutionary algorithm based on offset direction sampling and dual-layer competition (TSEAODS). TSEAODS employs an evolutionary algorithm framework, dividing its evolution process into rapid and precise convergence stages. In the rapid convergence stage, the proposed offset direction sampling strategy generates sample solutions to assist evolution, achieving rapid population convergence. In the precise convergence stage, the population is categorized into elite and non-elite layers based on non-dominated sorting. Through competition between different layers, two superior individuals are identified to further guide the population towards the Pareto-optimal front. Throughout the evolutionary process, the θ-dominance selection strategy (Yuan et al., 2016) is employed to maintain selection pressure and ensure diversity of population.

The main contributions of this paper are as follows.

1) An offset direction sampling strategy is proposed. A set of high-quality solutions is used to construct search directions. These directions are then randomly offset in angle and direction. Subsequently, a number of solutions are randomly sampled in all directions, and the non-dominated solutions are selected to guide population evolution. This strategy enhances the diversity of search directions, accelerates population convergence, and thoroughly explores the large-scale decision space.
2) A dual-layer competition strategy is proposed. Individuals of the population are divided into elite and non-elite layers based on non-dominated sorting. Competition among elite individuals identifies the global optimum, while competition between the elite and non-elite layers identifies the local optimum. Each non-elite individual learns from these optima in the decision space to prevent premature convergence.
3) The evolutionary process is divided into rapid and precise convergence stages. In the rapid convergence stage, the offset direction sampling strategy facilitates rapid population convergence. Due to the randomness of sampling, the further convergence becomes challenging after a certain point, thus necessitating the addition of the precise convergence stage. The dual-layer competition strategy is employed to exploit preferable solutions. The two-stage framework of the algorithm aims to balance search efficiency and quality.

The remainder of this paper is organized as follows. Section 2 outlines the background knowledge. The framework and core components of TSEAODS are described in Section 3. Section 4 details comparative experiments and analyzes their results. Finally, Section 5 provides conclusions.

2. Related work

Without loss of generality, the mathematical formulation for minimizing M objective functions in MOPs can be described as follows (Deb et al., 2002):

$$\begin{aligned} &\textit{minimize } \mathrm{F}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \cdots, f_M(\boldsymbol{x}))^T \\ &\text{s. t. } \boldsymbol{x} \in \Omega^d \end{aligned} \tag{1}$$

where $F(x)$ represents the objective vector to be optimized, $x \in \Omega^d$ represents that the decision vector belongs to the d-dimensional decision space, and M is the number of objectives. Specifically, when $d \geq 100$ and $1 < M \leq 3$, the MOPs in (1) are referred to as LSMOPs. The inherent conflicts among objectives often make it challenging to derive solutions that simultaneously satisfy all objectives. In this context, the Pareto-optimal set (PS) refers to solutions in the decision space, while the Pareto-optimal front (PF) corresponds to their mapping in the objective space (Li and Zhang, 2009).

According to the used strategies, existing LSMOEAs can be broadly classified into the following four categories.

1) Decision variable grouping-based LSMOEAs. The main idea of this category is to group decision variables based on their characteristics and then optimizing them separately. The pioneering multiobjective evolutionary algorithm (MOEA) was designed for addressing LSMOPs (Antonio and Coello, 2013), incorporated a random grouping strategy along with a cooperative coevolutionary framework. The algorithm partitioned decision variables into multiple subgroups, optimizing them cooperatively using a developed version of generalized differential evolution (Kukkonen and Lampinen, 2005). Later, a random dynamic grouping method combining multiobjective evolutionary algorithm based on decomposition (MOEA/D) was proposed (Song et al., 2016), which dynamically and randomly grouped decision variables to improve the efficiency of cooperative coevolution in optimizing subgroups. Cooperative coevolutionary large-scale multiobjective optimization method (Li and Wei, 2018) utilized a differential grouping technique and proposed a method for quickly identifying interdependencies, grouping interdependent variables based on their relationships to enhance the optimization capability of the cooperative coevolution framework. MOEA with decision variable analysis (MOEA/DVA) (Ma et al., 2016) applied a perturbation method to decision variables to ensure that the generated solutions exhibit dominance relationships. In a decision variable clustering based MOEA (Zhang et al., 2018), K-

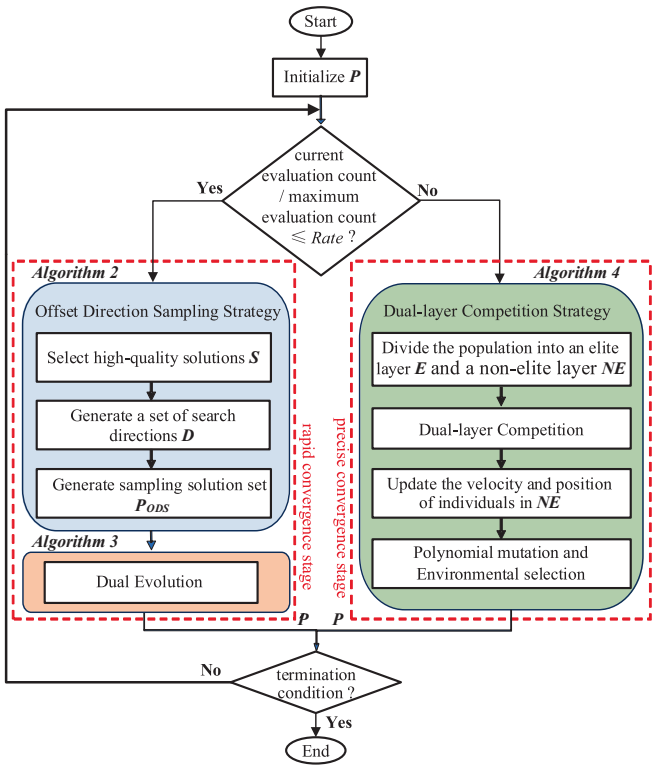


Fig. 1. The flowchart of TSEAODS.

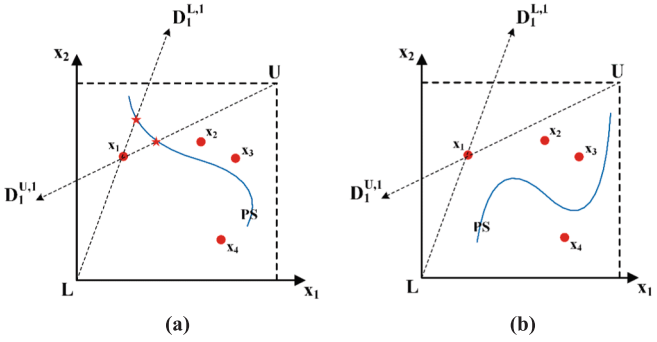


Fig. 2. Intersection of search directions with the PS. (a) Search directions intersect with the PS. (b) Search directions do not intersect with the PS.

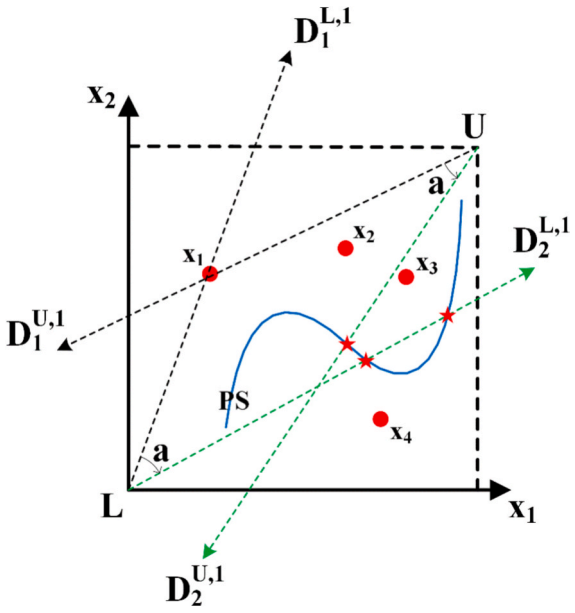


Fig. 3. Generation of offset directions.

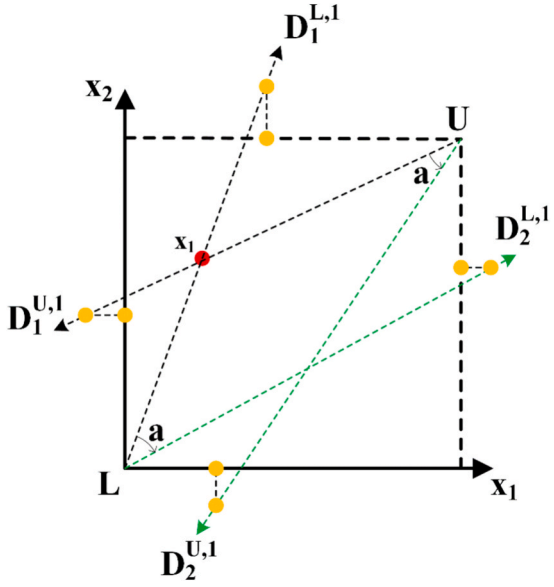


Fig. 4. Solution repair along the projection direction.

means clustering classified decision variables based on convergence and diversity. Additionally, a scalable small subpopulations-based covariance matrix adaptation evolution strategy (Chen et al., 2020) employed the same strategy of variable partitioning as MOEA/DVA. Each convergence-related variable group within a subpopulation was independently optimized, while diversity-related variables were shared across the entire subpopulation. Recently, a novel method formulated the analysis of decision variable as binary optimization problem (LERD) (He et al., 2024), enabling the generation of diverse grouping outcomes.

2) Problem transformation-based LSMOEAs. This category of LSMOEAs primarily focuses on reducing dimensionality, where conversion process is governed by the problem transformation function. The decision variables of weighted optimization framework (WOF) (Zille et al., 2018) were partitioned into several groups. The work (Liu et al., 2020) utilized a dynamic random grouping method, which reduces the decision space's search complexity by adjusting group sizes. Additionally, the work by Lin et al. (2015) was applied to refine the initial population and the weight vectors. However, in contrast to transformation-based methodology of WOF, the work by He et al. (2019) performed reconstruction by linking reference solutions within the decision space to corresponding weight vectors. As a result, the dimensionality of the initial decision space was compressed to match that of the predefined weight vectors. In a LSMOEA based on problem transformation (Sun and Jiang, 2024), the problem scale was effectively reduced using the opposition-based individual strategy and a dual transformation function.

3) Search direction-based LSMOEAs. This kind of LSMOEAs primarily focuses on producing offspring solutions through the design of effective search directions and systematic sampling within the decision space. In direction guided adaptive offspring generation method in a generic MOEA framework (DGEA) (He et al., 2022), diversity and convergence direction vectors were generated by combining non-dominated and dominated solutions, guided by the count of non-dominated solutions in the preselected population. In a memetic evolution system with statistical variable classification (Ge

et al., 2022), the top individual in each cluster was designated as the elite solution. Both the elite solution and other individuals were used to define sampling directions, enhancing the overall quality of individuals in the cluster. A LSMOEA assisted by directed sampling (LMOEADS) (Qin et al., 2021) constructed search directions and performed sampling by selecting the individual that was the nearest to the origin along each reference vector, guiding offspring generation and improving population convergence. Recently, a novel directed quick search-guided large-scale multiobjective evolution framework (Wu et al., 2024) presented a quick search-guided evolutionary framework to improve search efficiency and convergence quality. It achieved this through bidirectional vector-based sampling and quick search-guided reproduction strategies.

4) Novel search strategies-based LSMOEAs. This category of LSMOEAs employs novel search strategies to tackle LSMOPs. In a competitive swarm optimizer based efficient search (LMOCSO) (Tian et al., 2020), decision variables were not specially treated; instead, a new particle update strategy was used to enhance optimization performance. A comprehensive competitive swarm optimizer (Liu et al., 2022) utilized three competitive strategies to guide the solution search, enhancing the search capabilities of weaker particles. However, millions of evaluations were implemented to guarantee search efficiency. A fuzzy decision variables framework (FDV) (Yang et al., 2023) approach divided the evolutionary process into fuzzy and precise evolution. By fuzzifying the decision variables, it reduced the search space, facilitating rapid convergence of the population. A multi-stage competitive swarm optimization algorithm (Shang et al., 2025) enhanced global exploration through a fuzzy search strategy and adaptive dual-directional sampling. A novel update learning strategy was incorporated to accelerate convergence.

3. The proposed algorithm

3.1. The framework of TSEAODS

The framework of TSEAODS is outlined in Algorithm 1. The algorithm partitions the evolutionary process into two stages: a rapid convergence stage and a precise convergence stage.

Algorithm 1 The framework of TSEAODS

Input: N (population size), Rate (phase proportion),
N_V (the clustering number for reference vectors),
N_s (the number of random samples);
Output: P (final population);
1 $P \leftarrow$ Initialization (N);
2 **while** *termination criterion not fulfilled* **do**
3 **if** *the current evaluation count/the maximum evaluation count* $\leq$ Rate **then**
4 $P_{ODS} \leftarrow$ offset direction sampling strategy (N_V, N_s, V);
5 $P' \leftarrow$ dual evolution strategy (P_{ODS});
6 **else**
7 $P \leftarrow$ dual-layer competition strategy (P');
8 **end if**
9 **end while**
10 **return** P;

Initially, the population P is randomly initialized, and a set of reference vectors uniformly distributed across the objective space is generated. The proportion of the two stages, denoted as Rate, is determined. If the ratio of the current evaluation count to the maximum evaluation count is less than or equal to Rate, the algorithm enters the rapid convergence stage, employing the offset direction sampling strategy to optimize the population (lines 4–5). When the ratio exceeds Rate, the algorithm transitions to the precise convergence stage, applying the dual-layer competition strategy to refine the population (line 7). Fig. 1 presents the flowchart of TSEAODS. Algorithm 2 elaborates the offset direction sampling strategy, encompassing the selection of high-quality solutions, the definition of offset directions, and the process of random sampling. Algorithm 3 presents the dual evolution strategy, which leverages the final sampled solutions to facilitate population evolution. Algorithm 4 demonstrates the dual-layer competition strategy, categorizing the population into elite and non-elite layers. Competition results in the selection of two superior individuals, further guides the evolution of the population.

3.2. Offset of search directions

This section offers an introduction of the offset direction sampling strategy. Given that the search occurs within the decision space, it is essential to identify search directions within this space that intersect with PS. Subsequently, we sample along these identified search directions to acquire solutions that are closer to PS, thereby guiding the population's evolution. To accomplish this, we initially select solutions that are near the ideal point in the objective space, which are referred to as high-quality solutions. The selection of high-quality solutions is performed using the guide solution selection method as described in Qin et al. (2021). A set of uniformly distributed reference vectors V, is partitioned into $N_V - M$ clusters by the K-means. Next, to maintain the diversity of the population, M boundary vectors are added to V, where M is the number of objectives. Finally, N_V high-quality solutions are selected and added to the high-quality solution set S (lines 1–3 of Algorithm 2).

Algorithm 2 Offset Direction Sampling Strategy

Input: P_t (current population), V (a set of reference vectors),
N_V (the clustering number for reference vectors);
Output: P_{ODS} (final sampling solutions);
1 $V \leftarrow$ k-means (V, $N_V - M$);
2 $V \leftarrow$ Add M boundary reference vectors;
3 $S \leftarrow$ Select N_V high-quality solutions based on V;
4 $D_1 \leftarrow$ Establish two search directions as specified by (2) for each individual of the set S;
5 $a \leftarrow$ Generate a random angle between 0 and 90 degrees;
6 $r \leftarrow$ Generate a random direction; $//r = 1$ or -1
7 $R \leftarrow$ Calculate R according to (3);
8 $D_2 = R \times D_1$;
9 $D \leftarrow D_1 \cup D_2$;
10 $A \leftarrow$ Sample along the search directions in D;
11 $P_{ODS} \leftarrow$ Perform non-dominated sorting on all solutions in A and select the highest-ranked solutions;
12 **return** P_{ODS};

After obtaining the high-quality solution set S, two search directions are constructed for each solution x_j in S, starting from the lower bound L and upper bound U of the decision space, respectively. This results in a direction set D_1 (line 4). Compared to single-direction search, the bidirectional search aims to increase the probability of direction vectors intersecting with PS (Qin et al., 2021). Then, the direction vectors in D_1 are randomly offset in direction and angle to generate offset directions, forming a new direction set D_2. Finally, D_1 and D_2 are merged to obtain the final search direction set D (lines 5–9).

The search direction D_1 is defined as follows:

$$D_1^{L,i} = x_j - L$$

$$D_1^{U,i} = x_j - U \tag{2}$$

where $x_j, j = 1,2,\cdots,N_v$, represents all solutions in the set S. The search directions $D_1^{L,i}$ and $D_1^{U,i}$, where $i = 1,2,\cdots,N_v$, originate from L and U, respectively.

A total of $2\times N_v$ search directions are defined as $D_1^{L,1}, D_1^{U,1}, \cdots, D_1^{L,i}, D_1^{U,i}, \cdots, D_1^{L,N_v}, D_1^{U,N_v}$. For example, in Fig. 2, two lines are drawn starting from L and U, respectively, passing through the high-quality solution x_1, and generating two search directions $D_1^{L,1}$ and $D_1^{U,1}$.

Ideally, as shown in Fig. 2(a), $D_1^{L,1}$ and $D_1^{U,1}$ intersect with the PS (the blue curve), generating two intersection points. These points are considered optimal sampling solutions (red stars) and further guide the population towards the PS. However, in Fig. 2(b), $D_1^{L,1}$ and $D_1^{U,1}$ do not intersect with the PS, and the resulting sampled solutions fail to effectively guide the population.

In the first instance, generate a random angle a between 0 and 90 degrees, then determine the direction of the offset. Since the position of the PS cannot be predetermined, the offset direction needs to be chosen randomly, increasing the likelihood that the search direction intersects with the PS. The value $r = 1$ or -1 indicates the direction of the offset, with 1 representing counterclockwise and -1 representing clockwise. In a two-dimensional decision space, the transformation matrix R is first calculated, as shown in (3).

$$R = \begin{bmatrix} cos(r \times a) & -sin(r \times a) \\ sin(r \times a) & cos(r \times a) \end{bmatrix} \tag{3}$$

For an n-dimensional decision space, the transformation matrix is a corresponding n-dimensional matrix. The search direction is shifted as shown in (4).

$$D_2 = R \times D_1 \tag{4}$$

Fig. 3 illustrates an example where $D_1^{L,1}$ and $D_1^{U,1}$ are offset to obtain $D_2^{L,1}$ and $D_2^{U,1}$. Subsequently, $D_2^{L,1}$ and $D_2^{U,1}$ are stored in the direction set D_2. As shown in Fig. 3, four search directions, $D_1^{L,1}$, $D_1^{U,1}$, $D_2^{L,1}$ and $D_2^{U,1}$, are defined based on the quality solution x_1. Hence, for all solutions in S, $4 \times N_V$search directions will be defined. By comparing Figs. 2 and 3, it is observed that the introduction of offset directions increases the probability of intersection between all search directions and PS within the decision space. This is the most significant difference from the methods in other literature. Consequently, it becomes easier to sample solutions closer to PS, leading to faster convergence of the population towards the PS.

After defining all search directions, random sampling is performed on these direction vectors to generate a temporary solution set A. Then, all solutions in A have been ranked via non-dominated sorting, and the solutions in the highest level are selected and stored in the sampling solution set P_{ODS} (lines 10–11).

3.3. Random sampling

As shown in Fig. 3, for the search directions $D_1^{L,1}$, $D_1^{U,1}$, $D_2^{L,1}$ and $D_2^{U,1}$, N_s solutions are randomly sampled on each direction. For the high-quality solution x_1, $4\times N_s$ solutions are produced initially, and for all solutions in S, $4\times N_V\times N_s$ solutions are generated. The calculation method for generating solutions is given by (5):

$$x_m^{L,i} = L + s_m^{L,i} \cdot \frac{D_1^{L,i}}{\|D_1^{L,i}\|}$$

$$x_n^{U,i} = U + s_n^{U,i} \cdot \frac{D_1^{U,i}}{\|D_1^{U,i}\|}$$

$$x_j^{L,i} = L + s_j^{L,i} \cdot \frac{D_2^{L,i}}{\|D_2^{L,i}\|}$$

$$x_k^{U,i} = U + s_k^{U,i} \cdot \frac{D_2^{U,i}}{\|D_2^{U,i}\|} \tag{5}$$

where s is a non-negative coefficient, and the random variables $s_m^{L,i}$, $s_n^{U,i}$, $s_j^{L,i}$ and $s_k^{U,i}$ are uniformly distributed over the interval $[0, \|U-L\|]$, ensuring diversity in sampling. Additionally, $\|\cdot\|$ denotes the Euclidean distance. m, n, j and k correspond to the m-th, n-th, j-th and k-th solutions generated along the search directions $D_1^{L,1}$, $D_1^{U,1}$, $D_2^{L,1}$ and $D_2^{U,1}$, respectively.

Since the coefficient s can take any value within $[0, \|U-L\|]$, it may cause the generated solutions to exceed the upper or lower bounds of the decision space. Here, variables that violate the boundaries need to be clamped. As shown in Fig. 4, the yellow solid dots represent infeasible solutions that have exceeded the boundaries, which are clamped to the boundary. The experiments in Qin et al. (2021) show that this correction method is beneficial for exploring the boundaries of the decision space, thereby enhancing the exploration capability of algorithm.

3.4. Dual evolution

The motivation for the dual evolution strategy is to utilize sampled solutions to direct the current population towards generating offspring that are closer to PS, thus accelerating the convergence. Simultaneously, it is crucial to maintain a balance in convergence and diversity for the population. To address it, we adopt the dual reproduction strategy outlined in Qin et al. (2021), differing primarily in the application of the θ-dominance approach for environmental selection to balance population diversity and convergence. Algorithm 3 provides the pseudocode for the dual evolution strategy. Sampled solutions are utilized exclusively in the first evolution process to generate offspring solutions that are closer to PS, thereby accelerating the convergence. During the second evolution process, genetic operations are applied to the current population to enhance diversity and explore a broader solution space.

In the first evolution process, each individual in the current population is crossed with a randomly selected solution from the sampled solution set P_{ODS} to generate N candidate solutions. These solutions undergo polynomial mutation to form a temporary offspring population O_1 (lines 1–2). Then, the current population P_t, the newly generated temporary offspring population O_1, and the sampled solution set P_{ODS} are merged to form the combined population $C_1' = P_t \cup O_1 \cup P_{ODS}$. Environmental selection is performed on this population using the θ-dominance environmental selection strategy (lines 3–4). After that, the first selected population P_{t+1}' undergoes genetic operations, and then produce a new offspring population O_2(line 5). Finally, the combined population $C_1 = P_{t+1}' \cup O_2$, consisting of P_{t+1}' and O_2, undergoes the same environmental selection to produce the next generation population P_{t+1} (lines 6–8).

Algorithm 3 Dual Evolution Strategy

Input: P_t (current population), P_{ODS} (final sampling solutions), N (population size);
Output: P_{t+1} (next generation population);
1 For each individual in P_t, select a random solution from P_{ODS} for crossover, generate N candidate solutions;
2 O_1 ← mutate all candidate solutions by polynomial mutation;
3 $C_1' \leftarrow P_t \cup O_1 \cup P_{ODS}$;
4 P_{t+1}' ← Perform environmental selection on C_1';
5 O_2 ← Perform genetic operations on P_{t+1}';
6 $C_1 \leftarrow P_{t+1}' \cup O_2$;
7 P_{t+1} ← Perform environmental selection on C_1;
8 **return** P_{t+1};

3.5. Dual-layer competition

During the rapid convergence phase, due to the inherent randomness of the offset direction sampling strategy, the evolution may stagnate after reaching a certain point, making it difficult to continue approaching the PS. Therefore, a precise convergence phase is added. In this phase, a dual-layer competition strategy is employed to achieve precise evolution of the population, further guiding it towards convergence to the Pareto-optimal front. The details of this strategy are provided below, with the pseudocode shown in Algorithm 4.

Algorithm 4 Dual-layer Competition Strategy

Input: P'(population obtained from the fast convergence phase), N (population size);
Output: P (final population);
1 $P' \leftarrow$ Perform non-dominated sorting on P';
2 $E=\emptyset$, $NE=\emptyset$, $G=\emptyset$;
3 Assign the top $\lfloor N/2 \rfloor$ individuals from P' to set E and the remain individuals to set NE;
4 Initialize x_g, x_l, $B=\emptyset$;
5 Calculate convergence indicator $\mathrm{con}(x_e)$ and diversity indicator $\mathrm{div}(x_e)$ for all individuals in E;
6 **for** each individual x_{ne} in NE **do**
7 Randomly select two different individuals (x_e^1, x_e^2) from E, $x_g = \mathrm{gc}(x_e^1, x_e^2)$;
8 **for** each individual x_e in E **do**
9 **if** $lc(x_e, x_{ne}) == 1$ //x_e wins
10 Add x_e into B;
11 **end if**
12 **if** $|B| > 0$
13 Randomly select an individual from B to replace x_l;
14 **end if**
15 **end for**
16 **end for**
17 Update the velocity v_{ne} of individuals in NE according to (13);
18 Update the position x_{ne} of individuals in NE according to (14);
19 Perform polynomial mutation on all individuals;
20 $G \leftarrow E \cup NE$;
21 $P \leftarrow$ Perform environmental selection on the combined population P' and G;
22 **return** P;

Initially, non-dominated sorting is performed on population P'. Subsequently, three empty sets, denoted as E, NE, and G, are initialized. The top $\lfloor N/2 \rfloor$ individuals with the highest non-dominated ranks from P' are assigned to set E, referred to as the elite layer. The remaining individuals are assigned to set NE, designated as the non-elite layer. Each individual in sets E and NE is assigned an additional velocity attribute, v. An individual is randomly selected from E to serve as both the initial global best individual, x_g, and the local best individual, x_l (lines 1–4).

Next, all individuals in the elite layer compete to determine the global best individual, x_g. Calculate two performance indicators for all individuals in E, namely $\mathrm{con}(x_e)$ and $\mathrm{div}(x_e)$ (line 5). Here, the convergence performance of x_e, denoted as $\mathrm{con}(x_e)$, is quantified by the Euclidean distance between its objective value $F(x_e)$ and the ideal point z^*, as shown in (6).

$$\mathrm{con}(x_e) = \|F(x_e) - z^*\| \tag{6}$$

$\mathrm{div}(x_e)$ represents the diversity performance of x_e, defined by the shifted density estimation (Liu et al., 2022), as shown in (7).

$$\mathrm{div}(x_e) = \min_{x \in P \setminus \{x_e\}} \sqrt{\sum_{m}^{i=1} (max\{0, f_i(x) - f_i(x_e)\})^2} \tag{7}$$

Then, two different individuals (x_e^1, x_e^2) are randomly selected from E and compete based on the above two metrics (line 7). The winner of the competition replaces the current x_g. The calculation is shown in (8):

$$\mathrm{gc}(x_e^1, x_e^2) = \begin{cases} x_e^1, & if(r \le 0.5 \& \mathrm{con}(x_e^1) < \mathrm{con}(x_e^2)) \\ & or(r > 0.5 \& \mathrm{div}(x_e^1) > \mathrm{div}(x_e^2)) \\ x_e^2, & otherwise \end{cases} \tag{8}$$

where r is a random number following uniform distribution on the interval [0, 1].

Then, a competition is conducted among all individuals in the elite and non-elite layers to determine the local best individual x_l. Each individual x_{ne} in the set NE competes with each individual x_e in the set E, denoted as $\mathrm{lc}(x_e, x_{ne})$. Its calculation is shown in (9):

$$\mathrm{lc}(x_e, x_{ne}) = \begin{cases} 1, & if\ g(x_e|x_{ne}, z^*) < \|F(x_{ne}) - z^*\| \\ 0, & otherwise \end{cases} \tag{9}$$

where $z^* = (z_1^*, z_2^*, ..., z_m^*)$ is an ideal point. Calculating the exact ideal point is time-consuming. $g(x_e|x_{ne}, z^*)$ is similar to the boundary-crossing aggregation value based on penalties (Zhang and Li, 2007), and it is defined as follows:

$$g(x_e|x_{ne}, z^*) = d_1(x_e, F(x_{ne})) + \beta d_2(x_e, F(x_{ne})) \tag{10}$$

where the calculation of $d_1(x_e, F(x_{ne}))$ is shown in (11).

$$d_1(x_e, F(x_{ne})) = \frac{(F(x_e) - z^*) \cdot (F(x_{ne}) - z^*)}{\|F(x_e) - z^*\|} \tag{11}$$

The calculation of $d_2(x_e, F(x_{ne}))$ is shown in (12).

$$d_2(x_e, F(x_{ne})) = \|F(x_e) - z^* - \frac{(F(x_{ne}) - z^*) d_1(x_e, F(x_{ne}))}{\|(F(x_{ne}) - z^*)\|}\| \tag{12}$$

β is an adaptive value, set to $\beta = random(0, 1) + \frac{5t}{t_{max}}$. Here, t and t_{max} represent the current evaluation count and the maximum evaluation count, respectively. In (9), $lc(x_e, x_{ne}) = 1$ indicates that x_e wins the competition. Add the winning individual to set B. Then, randomly select an individual from B to replace the initial x_l (lines 8–15). Note that x_l is chosen from the winning individuals identified in (9). This approach adaptively and effectively guides the local search of the non-elite individuals.

Next, after determining x_g and x_l, all individuals in the non-elite layer learn from these two winning individuals. Their velocities v are updated according to (13):

$$v = r_1 v + r_2(x_l - x_{ne}) + r_3(x_g - x_{ne}) \tag{13}$$

where r_1, r_2, and r_3 are three randomly generated real-valued vectors within the range $[0, 1]^n$. The next step is to update their positions accordingly based on (14).

$$x = x + v \tag{14}$$

Polynomial mutation (Li et al., 2014) is applied to all individuals to prevent the population from getting trapped in local optima. Finally, these individuals are added to set G, and environmental selection is performed on the combined population of P' and G to obtain the final population P (lines 17–21). Here, the θ-dominance environmental selection strategy is still employed for population selection.

3.6. Computational complexity analysis

The computational complexity of the proposed TSEAODS in each generation is mainly determined by three components: the offset direction sampling strategy, the dual evolution strategy, and the dual-layer competition strategy. Let D represent the dimension of decision variable, N the population size, and M the number of objectives. The offset direction sampling strategy, which is responsible for constructing search directions and generating sampled solutions, has a complexity of $O(D^2)$. The dual evolution strategy, involving two rounds of genetic operations and environmental selection, has a complexity of $O(ND + N^2M)$. The dual-layer competition strategy includes non-dominated sorting, competition mechanisms, variation operations, and environmental selection, with a complexity of $O(ND + N^2M)$. As a result, the overall computational complexity of TSEAODS per generation is $O(D^2 + ND + N^2M)$. Compared with the baseline algorithm LMOEADS, whose complexity is $O(ND + N^2M)$, TSEAODS incurs an additional cost of $O(D^2)$. Despite the slightly higher computational overhead, the performance improvements observed in the experiments justify this additional cost.

4. Experiments

4.1. Benchmark problems

The benchmark test suite LSMOP comprises nine large-scale multi-objective problems: LSMOP1 − LSMOP9 (Cheng et al., 2017). LSMOP1-LSMOP4 feature Pareto-optimal fronts characterized by linear variable dependencies, including a linear unit hyperplane within the global Pareto-optimal front. In contrast, LSMOP5 − LSMOP8 exhibit nonlinear variable dependencies in their Pareto fronts, including a concave unit hypersphere. LSMOP9 involves nonlinear variable dependencies within its decision space and includes discontinuous line segments in its Pareto-optimal front.

Dimensions of objectives along with the decision variables are scalable in each LSMOP problem. To assess the effectiveness of TSEAODS, comparative experiments are performed on two and three-objective problems with decision variable counts of 500, 1000, 2000, and 5000, resulting in a total of 72 experiments. The experimental environment is Intel(R) Core(TM) i5-12400 @ 2.5 GHz and 24 GB RAM. All algorithms under comparison were implemented using PlatEMO (Tian et al., 2017) and MATLAB 2022. Each comparative algorithm runs independently 30 times on all test problems.

4.2. The compared algorithms and parameter settings

The algorithms compared in this study include LMOCSO (Tian et al., 2020), LMOEADS (Qin et al., 2021), FDV (Yang et al., 2023), DGEA (He et al., 2022), and LERD (He et al., 2024). LMOCSO and FDV employ novel search strategies, whereas LMOEADS and DGEA are direction-based algorithms. LERD integrates decision variable grouping with problem transformation techniques.

To ensure fairness across experiments, all algorithms were configured with an identical population size of 153 and the maximum number of function evaluations set at 100,000. Simulated binary crossover of TSEAODS is defined by a distribution index of 20 and a crossover probability of 0.9, while polynomial mutation applies the same distribution index and assigns a mutation probability of $1/d$. Here d represents the dimension of decision variables. In (5), the number of random samples generated along each search direction is set to 30. The initial clustering number for reference vectors is set to 10, while *Rate* is set at 0.7. Unique parameters for each comparative algorithm are set according to the values specified in their respective original papers.

4.3. Performance metrics

To assess the performance of TSEAODS comprehensively, the metrics of inverted generational distance (IGD) (Zhou et al., 2006) and hypervolume (HV) (Audet et al., 2021) are employed.

For a given set P, the IGD is calculated as follows:

$$IGD(P, P^*) = \frac{\sum_{x^* \in P^*} d(x^*, P)}{|P^*|} \tag{15}$$

where P^* denotes a collection of reference points that are uniformly distributed, x^* is reference solution, and $d(x^*, P)$ represents the minimum Euclidean distance from x^* in P^* to all solutions in set P. $|P^*|$ indicates the number of reference points in P^*. A lower IGD value corresponds to higher quality of the optimal solutions.

The definition of HV is as follows:

$$HV(Q) = \Lambda\left(\bigcup_{p \in Q} \{x | p \prec x \prec x_{ref}\}\right) \tag{16}$$

where Q denotes the approximate solution set located on the Pareto-optimal front, Λ is the Lebesgue measure, and x_{ref} is the reference point.

4.4. Analysis of Experimental results

To assess the differences in performance between TSEA-ODS and other comparative algorithms, the experiment uses the Wilcoxon rank-sum test with a significance level of 0.05 to evaluate whether the solution sets obtained by these algorithms on all test problems show significant differences (He and Yen, 2017). Symbols '+', '−' and '=' indicate that the comparative algorithms are statistically significantly better than, worse than, or comparable to TSEAODS, respectively. The best results are highlighted with a gray background and bold.

1) Experimental results and analysis of algorithms on two objectives. Tables 1 and 2 present the IGD and HV values for each algorithm on the LSMOP1-9 test problems, considering 500, 1000, 2000, and 5000 decision variables, each associated with two objectives. It is evident that TSEAODS achieves the best IGD and HV values across the majority of test problems, indicating that its offset direction sampling strategy and dual-layer competition strategy confer significant advantages in solving two-objective LSMOP problems. Under the constraint of limited function evaluations, TSEAODS obtains 36 superior IGD values and 28 superior HV values compared to LMOCSO. LMOCSO, constrained by its competition mechanism, necessitates millions of evaluations to achieve satisfactory results. FDV, which fuzzies decision variables to narrow the search space, demonstrates some effectiveness; however, achieving superior results also demands a substantial number of evaluations. Consequently, FDV does not surpass TSEAODS in terms of IGD and HV values. Additionally, TSEAODS demonstrates substantial advantages, outperforming LMOEADS in 22 IGD values and 20 HV values and surpassing DGEA in 34 IGD values and 28 HV values. Moreover, for LSMOP3 and LSMOP7 problems, all algorithms achieve an HV value of

Table 1
IGD results obtained by TSEAODS and five comparison algorithms on two objectives with different dimensions.

Problem	M	D	LMOCSO	LMOEADS	FDV	DGEA	LERD	TSEAODS
LSMOP1	2	500	6.5309e-1 (8.01e-2) -	2.9915e-1 (1.18e-2) -	1.6108e+0 (2.48e-1) -	3.3885e-1 (4.59e-2) -	8.8032e-1 (8.50e-2) -	**2.7593e-1 (4.98e-3)**
		1000	1.4199e+0 (1.45e-1) -	3.2225e-1 (1.17e-2) -	3.1297e+0 (1.98e-1) -	3.7980e-1 (1.13e-1) -	1.3481e+0 (1.41e-1) -	**2.9603e-1 (1.03e-2)**
		2000	1.9465e+0 (1.25e-1) -	3.2937e-1 (9.72e-3) -	4.4271e+0 (1.87e-1) -	4.1710e-1 (1.17e-1) -	1.5354e+0 (1.54e-1) -	**3.0701e-1 (1.21e-2)**
		5000	2.2396e+0 (1.80e-1) -	3.3984e-1 (7.21e-3) -	6.1351e+0 (1.92e-1) -	3.7267e-1 (6.39e-2) -	1.5716e+0 (1.88e-1) -	**3.1194e-1 (1.18e-2)**
LSMOP2	2	500	4.2776e-2 (6.61e-4) -	1.3044e-2 (3.74e-4) -	6.1719e-2 (6.58e-4) -	1.3933e-2 (5.15e-4) -	3.5017e-2 (6.38e-3) -	**1.1588e-2 (1.63e-4)**
		1000	2.3723e-2 (4.14e-4) -	7.8209e-3 (2.28e-4) -	3.8091e-2 (2.19e-4) -	8.5143e-3 (4.95e-4) -	2.0318e-2 (1.43e-3) -	**6.7022e-3 (2.65e-4)**
		2000	1.3359e-2 (2.21e-4) -	5.0196e-3 (1.04e-4) -	2.1096e-2 (4.72e-5) -	5.7032e-3 (3.74e-4) -	1.2004e-2 (5.46e-4) -	**4.1686e-3 (6.01e-5)**
		5000	6.6065e-3 (7.59e-5) -	3.5360e-3 (1.23e-4) -	1.0161e-2 (7.02e-5) -	4.0322e-3 (1.74e-4) -	6.2801e-3 (2.18e-4) -	**2.8475e-3 (2.86e-5)**
LSMOP3	2	500	2.0135e+0 (2.59e+0) -	1.5656e+0 (7.62e-4) +	1.7001e+1 (1.37e+0) -	**1.4836e+0 (6.30e-1)** +	6.7582e+0 (6.28e+0) -	1.5679e+0 (9.67e-4)
		1000	4.6825e+0 (3.04e+0) -	**1.5735e+0 (1.03e-3)** +	2.0022e+1 (9.42e-1) -	1.6338e+0 (6.30e-1) -	1.2609e+1 (4.12e+0) -	1.5760e+0 (3.61e-4)
		2000	1.0112e+1 (5.51e+0) -	**1.5770e+0 (7.02e-4)** +	2.3551e+1 (5.04e-1) -	2.4654e+0 (2.02e+0) -	1.3904e+1 (1.90e+0) -	1.5799e+0 (1.66e-4)
		5000	1.8518e+1 (3.59e+0) -	**1.5787e+0 (5.24e-4)** +	2.7986e+1 (6.21e-1) -	3.4116e+0 (2.63e+0) =	1.5026e+1 (1.29e+0) -	1.5825e+0 (2.32e-3)
LSMOP4	2	500	8.5771e-2 (1.45e-3) -	4.0212e-2 (6.57e-4) -	9.3735e-2 (1.54e-3) -	4.6250e-2 (2.96e-3) -	5.7497e-2 (5.36e-3) -	**3.8722e-2 (2.10e-4)**
		1000	5.1079e-2 (3.81e-4) -	2.1957e-2 (4.31e-4) -	6.2982e-2 (8.40e-4) -	2.5705e-2 (1.62e-3) -	3.4423e-2 (2.77e-3) -	**2.0595e-2 (7.72e-5)**
		2000	2.9430e-2 (1.72e-4) -	1.2064e-2 (2.45e-4) -	3.9537e-2 (4.02e-4) -	1.4257e-2 (7.47e-4) -	1.9829e-2 (1.84e-3) -	**1.1157e-2 (1.13e-4)**
		5000	1.4341e-2 (4.93e-5) -	6.0891e-3 (1.23e-4) -	2.0244e-2 (1.57e-4) -	7.2317e-3 (4.02e-4) -	9.2354e-3 (4.62e-4) -	**5.4383e-3 (9.64e-5)**
LSMOP5	2	500	1.4808e+0 (2.77e-1) -	**7.4204e-1 (6.26e-5)** +	2.9301e+0 (1.63e+0) -	4.4980e+0 (1.03e+0) -	2.0492e+0 (9.76e-1) -	7.4209e-1 (8.41e-6)
		1000	3.4697e+0 (3.59e-1) -	**7.4202e-1 (7.35e-5)** +	7.0895e+0 (1.22e+0) -	4.7108e+0 (1.58e+0) -	4.5425e+0 (1.05e+0) -	7.4210e-1 (1.42e-5)
		2000	4.8461e+0 (6.49e-1) -	**7.4202e-1 (7.08e-5)** +	9.7617e+0 (8.98e-1) -	4.6461e+0 (1.46e+0) -	5.1950e+0 (8.84e-1) -	7.4209e-1 (8.49e-6)
		5000	5.4954e+0 (5.02e-1) -	**7.4198e-1 (1.08e-4)** +	1.4678e+1 (5.24e-1) -	3.7733e+0 (2.09e+0) -	5.4581e+0 (1.10e+0) -	7.4210e-1 (1.35e-5)
LSMOP6	2	500	7.8430e-1 (6.76e-3) -	3.2068e-1 (9.65e-3) -	8.0643e-1 (1.37e-3) -	6.0597e-1 (2.21e-1) -	7.5807e-1 (2.79e-2) -	**2.1689e-1 (6.94e-2)**
		1000	7.6744e-1 (4.47e-3) -	3.3013e-1 (8.00e-2) -	7.7429e-1 (7.57e-4) -	6.1876e-1 (2.04e-1) -	7.4146e-1 (4.12e-2) -	**2.0010e-1 (3.15e-2)**
		2000	7.5285e-1 (1.36e-2) -	3.0870e-1 (1.56e-4) -	7.5738e-1 (3.95e-13) -	5.8499e-1 (2.15e-1) -	7.5027e-1 (1.41e-2) -	**1.8257e-1 (1.37e-2)**
		5000	7.4752e-1 (1.83e-4) -	3.0700e-1 (3.50e-5) -	7.4774e-1 (3.09e-13) -	5.9121e-1 (2.13e-1) -	7.4673e-1 (3.14e-4) -	**1.7352e-1 (1.83e-2)**
LSMOP7	2	500	1.0524e+2 (6.03e+1) -	**1.5033e+0 (1.15e-3)** =	9.5065e+1 (2.12e+1) -	2.8549e+3 (1.47e+3) -	3.7069e+0 (6.11e-1) -	1.5035e+0 (4.53e-4)
		1000	1.8370e+3 (7.35e+2) -	1.5110e+0 (4.49e-4) =	1.7401e+3 (3.02e+2) -	3.2625e+3 (2.10e+3) -	6.1978e+0 (7.90e-1) -	**1.5107e+0 (3.38e-4)**
		2000	4.7728e+3 (1.41e+3) -	1.5147e+0 (4.93e-4) -	9.0130e+3 (1.18e+3) -	3.9829e+3 (2.87e+3) -	6.6872e+1 (6.61e+1) -	**1.5141e+0 (2.06e-4)**
		5000	6.9746e+3 (1.78e+3) -	1.5163e+0 (1.75e-4) -	2.6529e+4 (1.50e+3) -	2.8266e+3 (3.00e+3) -	1.6413e+3 (1.27e+3) -	**1.5160e+0 (3.15e-4)**
LSMOP8	2	500	1.2351e+0 (2.59e-1) -	6.5225e-1 (7.26e-2) =	2.2901e+0 (3.56e-1) -	1.9727e+0 (1.10e+0) -	**5.4128e-1 (1.77e-1)** +	6.3766e-1 (2.56e-2)
		1000	2.8794e+0 (3.08e-1) -	**7.4184e-1 (7.32e-4)** +	5.3616e+0 (3.11e-1) -	1.6798e+0 (1.06e+0) -	1.5699e+0 (3.92e-1) -	7.4209e-1 (7.59e-6)
		2000	3.8662e+0 (3.59e-1) -	**7.4201e-1 (1.22e-4)** +	8.3536e+0 (3.67e-1) -	2.3036e+0 (1.49e+0) -	2.0933e+0 (6.81e-1) -	7.4210e-1 (1.97e-5)
		5000	4.3389e+0 (4.47e-1) -	**7.4205e-1 (3.56e-5)** +	1.1810e+1 (1.96e-1) -	1.6814e+0 (1.12e+0) -	2.6480e+0 (6.20e-1) -	7.4209e-1 (2.97e-6)
LSMOP9	2	500	6.0874e-1 (1.21e-1) -	6.8694e-1 (6.09e-2) -	8.5485e-1 (4.99e-2) -	3.5356e+0 (1.06e+0) -	8.9942e-1 (7.75e-2) -	**6.4554e-2 (1.14e-3)**
		1000	2.3753e+0 (5.22e-1) -	5.7947e-1 (8.89e-2) -	1.6268e+0 (2.78e-1) -	7.9690e+0 (1.95e+0) -	1.0207e+0 (1.16e-1) -	**3.4179e-2 (4.53e-3)**
		2000	6.5065e+0 (1.78e+0) -	5.9369e-1 (3.50e-2) -	7.7989e+0 (1.19e+0) -	1.2112e+1 (4.57e+0) -	3.9548e+0 (2.16e+0) -	**2.1708e-2 (1.01e-2)**
		5000	1.1014e+1 (2.58e+0) -	5.8104e-1 (4.55e-2) -	2.1638e+1 (1.22e+0) -	1.5505e+1 (4.05e+0) -	7.7727e+0 (1.94e+0) -	**1.2527e-2 (3.96e-3)**
+/-/=			0/36/0	11/22/3	0/36/0	1/34/1	1/35/0	

0, highlighting the challenge that multimodal problems pose for algorithmic convergence.

Fig. 5 illustrates the IGD value variation curves of TSEAODS and the compared algorithms in the context of two-objective 5000-dimensional test problems on LSMOP1, LSMOP2, LSMOP4, LSMOP6, and LSMOP9. It is evident that TSEAODS exhibits notable performance advantages in the linear problems LSMOP1, LSMOP2, and LSMOP4, the convex problem LSMOP6, and the discontinuous problem LSMOP9. Among these, TSEAODS demonstrates the highest performance in LSMOP9. It is because our two-stage algorithm framework, along with the offset direction sampling strategy and dual-layer competition strategy, significantly enhances convergence speed and overall performance.

To further evaluate the performance of TSEAODS and the compared algorithms in the objective space for two-objective LSMOP problems, Fig. 6 presents a comparison of the results with the true PF on LSMOP1, LSMOP2, LSMOP4, LSMOP6, and LSMOP9 for two-objective 5000-dimensional test problems. Fig. 6 illustrates that TSEAODS converges closely and uniformly to the true PF on LSMOP2, LSMOP4, and LSMOP9. In the case of LSMOP1, TSEAODS and LMOEADS show a few solutions converging to the true PF, while LMOCSO, FDV, and LERD demonstrate that results are distant from the true PF, with no solutions converging to it. Overall, results of TSEAODS are nearer to and more uniformly distributed across the true PF compared to other algorithms. In summary, the proposed strategy provides TSEAODS with a substantial advantage.

2) Experimental results and analysis of algorithms on three objectives. Tables 3 and 4 list the IGD and HV values for each algorithm for three-objective LSMOP1-9 test problems, considering 500, 1000, 2000, and 5000 decision variables. The results show that TSEAODS achieves the best IGD and HV values on most test problems, demonstrating superior performance relative to two-objective LSMOP problems and further highlighting its overall advantage. For FDV, DGEA, and LERD, IGD values do not surpass those of TSEAODS on any problem; regarding HV values, they achieve values close to TSEAODS in only 1, 4 and 1 cases, respectively. LMOCSO outperforms TSEAODS in terms of HV values in only two problems. LMOEADS, which also uses the direction sampling

Table 2
HV results obtained by TSEAODS and five comparison algorithms on two objectives with different dimensions.

Problem	M	D	LMOCSO	LMOEADS	FDV	DGEA	LERD	TSEAODS
LSMOP1	2	500	5.0806e-2 (2.86e-2) -	2.4317e-1 (9.91e-3) -	0.000e+0 (0.00e+0) -	2.2114e-1 (4.62e-2) -	2.7974e-3 (5.44e-3) -	**2.5796e-1 (1.67e-2)**
		1000	0.000e+0 (0.00e+0) -	2.3328e-1 (1.24e-2) -	0.000e+0 (0.00e+0) -	2.0767e-1 (6.22e-2) -	0.000e+0 (0.00e+0) -	**2.5182e-1 (2.13e-2)**
		2000	0.000e+0 (0.00e+0) -	2.2717e-1 (1.03e-2) -	0.000e+0 (0.00e+0) -	1.8113e-1 (7.30e-2) -	0.000e+0 (0.00e+0) -	**2.4760e-1 (1.95e-2)**
		5000	0.000e+0 (0.00e+0) -	2.1992e-1 (7.34e-3) -	0.000e+0 (0.00e+0) -	2.0277e-1 (4.85e-2) -	0.000e+0 (0.00e+0) -	**2.4661e-1 (1.54e-2)**
LSMOP2	2	500	5.3019e-1 (8.43e-4) -	5.6820e-1 (5.19e-4) -	5.0597e-1 (7.57e-4) -	5.6754e-1 (6.36e-4) -	5.4055e-1 (8.55e-3) -	**5.7096e-1 (2.05e-4)**
		1000	5.5498e-1 (5.88e-4) -	5.7480e-1 (3.14e-4) -	5.3590e-1 (3.25e-4) -	5.7435e-1 (6.64e-4) -	5.5953e-1 (1.83e-3) -	**5.7701e-1 (3.55e-4)**
		2000	5.6828e-1 (2.84e-4) -	5.7859e-1 (1.93e-4) -	5.5809e-1 (1.16e-4) -	5.7811e-1 (5.34e-4) -	5.7006e-1 (6.82e-4) -	**5.8039e-1 (1.88e-4)**
		5000	5.7693e-1 (9.94e-5) -	5.8097e-1 (2.56e-4) -	5.7201e-1 (1.29e-4) -	5.8072e-1 (2.39e-4) -	5.7737e-1 (2.86e-4) -	**5.8248e-1 (5.92e-5)**
LSMOP3	2	500	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	**0.000e+0 (0.00e+0)**
		1000	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	**0.000e+0 (0.00e+0)**
		2000	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	**0.000e+0 (0.00e+0)**
		5000	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	**0.000e+0 (0.00e+0)**
LSMOP4	2	500	4.7806e-1 (1.68e-3) -	5.3294e-1 (9.21e-4) -	4.6716e-1 (1.74e-3) -	5.2538e-1 (3.75e-3) -	5.1204e-1 (6.56e-3) -	**5.3585e-1 (2.45e-4)**
		1000	5.1937e-1 (4.68e-4) -	5.5687e-1 (6.29e-4) -	5.0443e-1 (1.07e-3) -	5.5198e-1 (2.17e-3) -	5.4075e-1 (3.69e-3) -	**5.5947e-1 (1.01e-4)**
		2000	5.4694e-1 (2.22e-4) -	5.6963e-1 (3.22e-4) -	5.3397e-1 (5.10e-4) -	5.6704e-1 (9.76e-4) -	5.5990e-1 (2.48e-3) -	**5.7132e-1 (2.71e-4)**
		5000	5.6695e-1 (6.18e-5) -	5.7734e-1 (1.89e-4) -	5.5915e-1 (1.78e-4) -	5.7620e-1 (5.40e-4) -	5.7356e-1 (6.05e-4) -	**5.7874e-1 (1.28e-4)**
LSMOP5	2	500	0.000e+0 (0.00e+0) -	**9.0904e-2 (6.04e-6)** +	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	1.3636e-2 (3.33e-2) -	9.0901e-2 (2.08e-5)
		1000	0.000e+0 (0.00e+0) -	**9.0904e-2 (6.14e-6)** =	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	9.0896e-2 (3.59e-5)
		2000	0.000e+0 (0.00e+0) -	**9.0902e-2 (6.95e-6)** =	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	9.0900e-2 (2.02e-5)
		5000	0.000e+0 (0.00e+0) -	**9.0900e-2 (1.03e-5)** =	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	9.0895e-2 (3.30e-5)
LSMOP6	2	500	1.3471e-2 (9.53e-3) -	4.3019e-2 (8.71e-3) -	0.0000e+0 (0.00e+0) -	4.5013e-2 (2.54e-2) -	7.8261e-3 (6.74e-3) -	**1.4422e-1 (1.64e-2)**
		1000	3.9447e-2 (4.84e-3) -	6.8657e-2 (1.20e-2) -	2.9152e-2 (1.27e-3) -	6.2274e-2 (1.44e-2) -	4.4548e-2 (2.60e-3) -	**1.4975e-1 (1.65e-2)**
		2000	6.1961e-2 (1.92e-3) -	8.9626e-2 (6.58e-4) -	5.9011e-2 (2.49e-14) -	8.1112e-2 (1.76e-2) -	6.6728e-2 (1.85e-3) -	**1.6206e-1 (1.31e-2)**
		5000	7.8646e-2 (3.89e-4) -	1.0083e-1 (1.83e-4) -	7.8177e-2 (2.51e-14) -	8.9065e-2 (8.14e-3) -	8.0343e-2 (6.77e-4) -	**1.7108e-1 (1.00e-2)**
LSMOP7	2	500	**0.000e+0 (0.00e+0)** =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0)
		1000	**0.000e+0 (0.00e+0)** =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0)
		2000	**0.000e+0 (0.00e+0)** =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0)
		5000	**0.000e+0 (0.00e+0)** =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0)
LSMOP8	2	500	0.000e+0 (0.00e+0) -	9.0906e-2 (6.46e-6) =	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	4.3642e-2 (3.89e-2) -	**9.0907e-2 (3.44e-6)**
		1000	0.000e+0 (0.00e+0) -	**9.0908e-2 (1.89e-6)** +	0.000e+0 (0.00e+0) -	7.1446e-3 (2.18e-2) -	0.000e+0 (0.00e+0) -	9.0900e-2 (2.16e-5)
		2000	0.000e+0 (0.00e+0) -	**9.0909e-2 (5.58e-7)** +	0.000e+0 (0.00e+0) -	8.1588e-3 (2.11e-2) -	0.000e+0 (0.00e+0) -	9.0893e-2 (5.13e-5)
		5000	0.000e+0 (0.00e+0) -	**9.0908e-2 (3.28e-6)** =	0.000e+0 (0.00e+0) -	7.3932e-3 (1.84e-2) -	0.000e+0 (0.00e+0) -	9.0905e-2 (6.86e-6)
LSMOP9	2	500	6.1552e-2 (1.75e-2) -	1.1100e-1 (1.20e-2) -	5.5820e-2 (1.01e-2) -	2.6465e-3 (1.18e-2) -	6.7890e-2 (1.96e-2) -	**2.1241e-1 (4.39e-4)**
		1000	0.000e+0 (0.00e+0) -	1.3206e-1 (1.13e-2) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	3.9410e-2 (2.11e-2) -	**2.2537e-1 (1.81e-3)**
		2000	0.000e+0 (0.00e+0) -	1.3336e-1 (8.49e-3) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	1.9175e-3 (8.58e-3) -	**2.3183e-1 (3.79e-3)**
		5000	0.000e+0 (0.00e+0) -	1.3625e-1 (8.58e-3) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	**2.3671e-1 (1.84e-3)**
+/-/=			0/28/8	3/20/13	0/28/8	0/28/8	0/28/8	

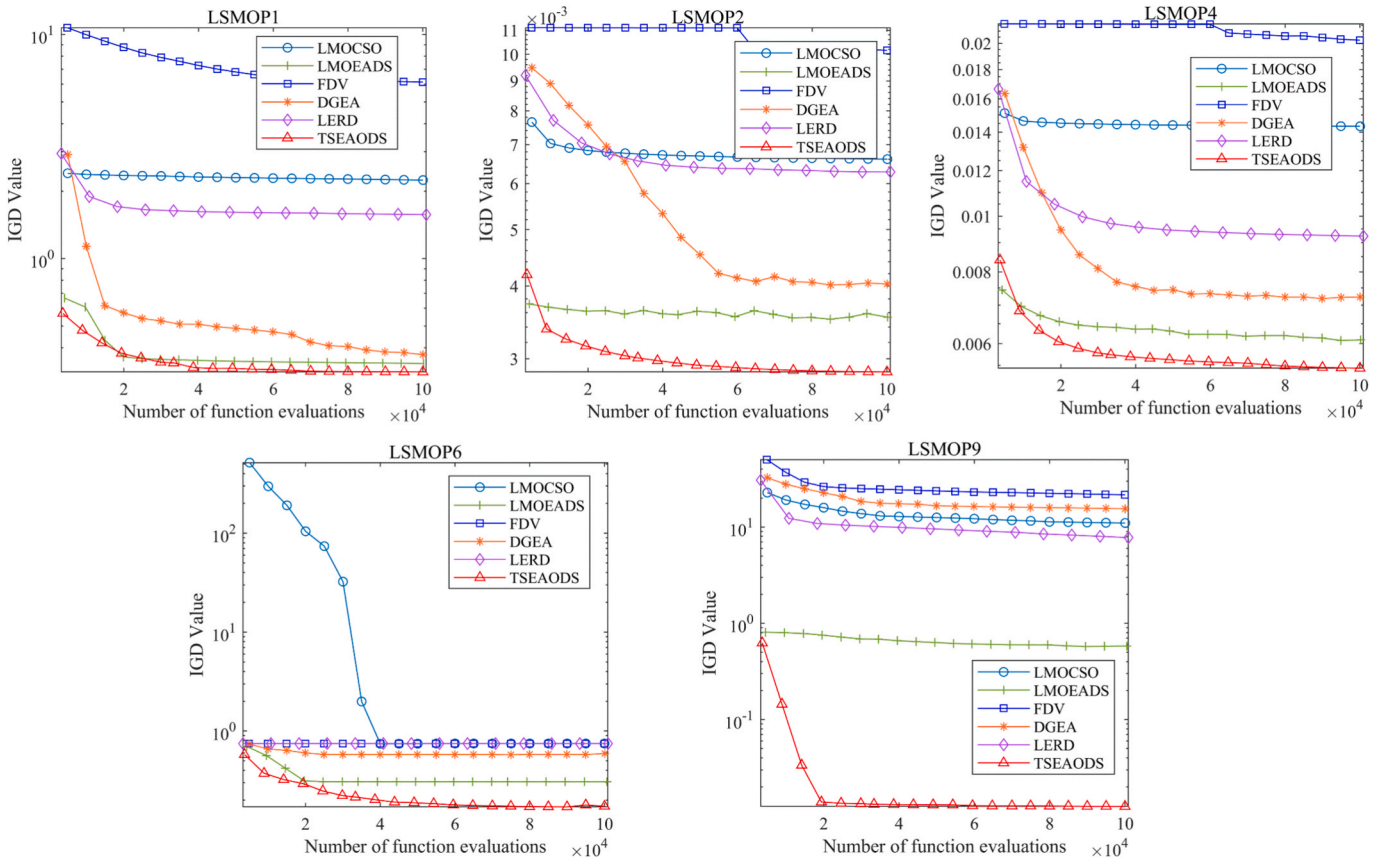


Fig. 5. Curves of the mean IGD value obtained by LMOCSO, LMOEADS, FDV, DGEA, LERD and TSEAODS on two-objective LSMOP1, LSMOP2, LSMOP4, LSMOP6 and LSMOP9 with 5000 decision variables.

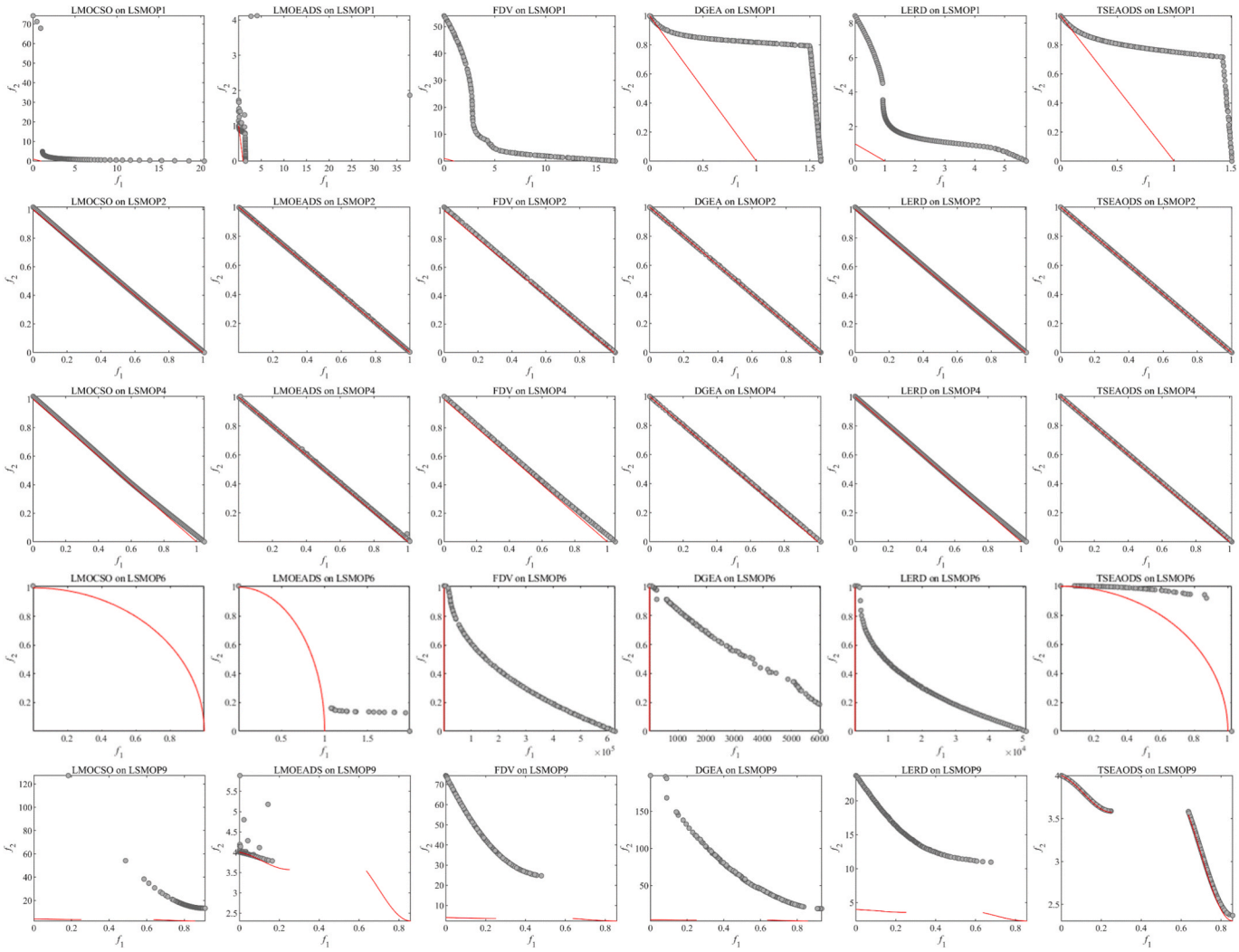


Fig. 6. The final solution sets obtained by LMOCSO, LMOEADS, FDV, DGEA, LERD and TSEAODS on two-objective LSMOP1, LSMOP2, LSMOP4, LSMOP6 and LSMOP9 with 5000 decision variables.

Table 3
IGD results obtained by TSEAODS and five comparison algorithms on three objectives with different dimensions.

Problem	M	D	LMOCSO	LMOEADS	FDV	DGEA	LERD	TSEAODS
LSMOP1	3	500	1.3168e+0 (1.06e-1) -	5.1020e-1 (4.13e-2) -	2.7331e+0 (2.32e-1) -	6.1494e-1 (1.05e-1) -	1.1418e+0 (2.01e-1) -	**3.6293e-1 (2.88e-2)**
		1000	1.5809e+0 (9.55e-2) -	5.6150e-1 (5.04e-2) -	4.4109e+0 (3.66e-1) -	7.1455e-1 (1.61e-1) -	1.2937e+0 (1.83e-1) -	**3.6489e-1 (3.71e-2)**
		2000	1.6758e+0 (8.25e-2) -	5.9318e-1 (3.87e-2) -	7.4228e+0 (5.43e-1) -	7.0179e-1 (1.00e-1) -	1.4476e+0 (2.32e-1) -	**3.6359e-1 (3.11e-2)**
		5000	1.7430e+0 (9.62e-2) -	6.0228e-1 (5.05e-2) -	9.3026e+0 (4.09e-1) -	7.1939e-1 (1.51e-1) -	1.4285e+0 (9.33e-2) -	**3.6662e-1 (3.15e-2)**
LSMOP2	3	500	5.0244e-2 (4.42e-4) -	4.4325e-2 (1.15e-3) -	7.5291e-2 (2.22e-3) -	6.0574e-2 (2.29e-3) -	6.1519e-2 (1.33e-3) -	**3.8316e-2 (5.70e-4)**
		1000	3.9889e-2 (2.87e-4) -	3.7654e-2 (1.14e-3) -	5.8948e-2 (3.02e-3) -	5.2291e-2 (2.83e-3) -	4.4503e-2 (4.03e-4) -	**3.4303e-2 (2.57e-4)**
		2000	3.5187e-2 (1.57e-4) -	3.4551e-2 (1.10e-3) -	5.0738e-2 (2.78e-3) -	4.8485e-2 (2.45e-3) -	3.6940e-2 (1.42e-4) -	**3.2615e-2 (1.28e-4)**
		5000	3.2636e-2 (3.97e-5) -	3.3158e-2 (1.11e-3) -	4.8041e-2 (2.56e-3) -	4.6591e-2 (1.87e-3) -	3.3337e-2 (1.09e-4) -	**3.1671e-2 (5.36e-5)**
LSMOP3	3	500	1.0657e+1 (2.30e+0) -	**8.6015e-1 (1.16e-3)** +	1.0615e+1 (2.41e+0) -	1.4355e+0 (1.88e+0) =	9.4288e+0 (4.71e-1) -	8.6084e-1 (5.31e-4)
		1000	1.3332e+1 (3.30e+0) -	**8.5998e-1 (2.32e-3)** +	1.4959e+1 (1.91e+0) -	1.2239e+0 (9.06e-1) =	9.8006e+0 (3.62e-1) -	8.6072e-1 (2.64e-4)
		2000	1.4528e+1 (2.70e+0) -	**8.6099e-1 (1.18e-3)** =	1.9967e+1 (2.55e+0) -	1.2768e+0 (1.47e+0) =	1.0032e+1 (3.49e-1) -	8.6062e-1 (4.98e-5)
		5000	1.4873e+1 (3.57e+0) -	8.6090e-1 (1.24e-3) -	2.2460e+1 (4.16e+0) -	1.7195e+0 (1.96e+0) -	1.0067e+1 (2.42e-1) -	**8.6090e-1 (9.76e-4)**
LSMOP4	3	500	1.4740e-1 (1.65e-3) -	1.0607e-1 (2.42e-3) -	1.9380e-1 (3.81e-3) -	1.2192e-1 (3.87e-3) -	1.4530e-1 (4.59e-3) -	**8.7194e-2 (2.81e-3)**
		1000	9.1277e-2 (5.70e-4) -	6.7515e-2 (1.38e-3) -	1.2234e-1 (3.06e-3) -	7.9119e-2 (2.46e-3) -	9.8505e-2 (2.90e-3) -	**5.8156e-2 (1.51e-3)**
		2000	5.8489e-2 (2.37e-4) -	4.8527e-2 (1.22e-3) -	8.1531e-2 (2.48e-3) -	5.9329e-2 (2.59e-3) -	6.3552e-2 (1.46e-3) -	**4.2889e-2 (5.80e-4)**
		5000	3.9393e-2 (1.50e-4) -	3.7313e-2 (8.52e-4) -	5.6055e-2 (2.78e-3) -	4.9345e-2 (2.00e-3) -	4.1577e-2 (3.71e-4) -	**3.4989e-2 (2.92e-4)**
LSMOP5	3	500	2.7968e+0 (2.62e-1) -	5.2700e-1 (1.48e-2) -	5.8414e+0 (6.13e-1) -	6.9931e-1 (4.04e-1) -	2.4211e+0 (9.41e-1) -	**4.6662e-1 (1.02e-2)**
		1000	3.4384e+0 (2.38e-1) -	5.3475e-1 (5.97e-3) -	1.0776e+1 (8.15e-1) -	5.5737e-1 (8.81e-2) -	3.2012e+0 (6.26e-1) -	**4.8977e-1 (6.31e-3)**
		2000	3.6677e+0 (1.72e-1) -	5.3716e-1 (4.88e-3) -	1.7464e+1 (6.02e-1) -	7.8068e-1 (4.71e-1) -	3.2860e+0 (2.74e-1) -	**5.0355e-1 (3.40e-3)**
		5000	3.8869e+0 (2.29e-1) -	5.3918e-1 (1.43e-3) -	2.0566e+1 (4.78e-1) -	6.9315e-1 (3.87e-1) -	3.0805e+0 (3.91e-1) -	**5.1513e-1 (1.29e-3)**
LSMOP6	3	500	8.7050e+1 (7.18e+1) -	7.7141e-1 (6.16e-2) -	1.3122e+2 (8.35e+1) -	1.3346e+1 (2.36e+1) -	4.7126e+0 (2.55e+0) -	**6.5930e-1 (4.53e-2)**
		1000	3.5740e+2 (1.50e+2) -	7.7223e-1 (4.03e-2) -	2.0322e+3 (7.57e+2) -	4.1432e+1 (6.54e+1) -	3.0651e+1 (5.09e+1) -	**6.6716e-1 (7.23e-2)**
		2000	8.4264e+2 (2.28e+2) -	7.8110e-1 (2.95e-2) -	6.4983e+3 (1.57e+3) -	1.2859e+2 (2.86e+2) -	1.5260e+2 (2.19e+2) -	**6.7115e-1 (6.70e-2)**
		5000	1.0622e+3 (4.96e+2) -	7.6748e-1 (3.38e-2) -	2.4787e+4 (4.62e+3) -	1.2458e+2 (2.11e+2) -	4.7608e+2 (3.31e+2) -	**6.9238e-1 (7.18e-2)**
LSMOP7	3	500	1.0057e+0 (8.12e-2) -	8.8500e-1 (3.97e-3) -	1.2764e+0 (1.84e-2) -	9.9388e-1 (9.44e-2) -	1.0400e+0 (9.93e-2) -	**7.9611e-1 (1.82e-2)**
		1000	9.9256e-1 (5.80e-2) -	8.5531e-1 (1.18e-2) -	1.1079e+0 (5.16e-3) -	9.6146e-1 (6.57e-2) -	1.0225e+0 (3.40e-2) -	**7.7988e-1 (1.24e-1)**
		2000	9.6888e-1 (6.38e-2) -	8.4537e-1 (9.40e-4) -	1.0209e+0 (1.30e-3) -	9.4627e-1 (4.91e-2) -	9.8655e-1 (5.34e-3) -	**7.2421e-1 (6.82e-2)**
		5000	9.3123e-1 (6.72e-2) -	8.3886e-1 (3.44e-4) -	9.7287e-1 (3.63e-4) -	9.3937e-1 (3.68e-2) -	9.6165e-1 (1.58e-3) -	**6.9807e-1 (4.06e-3)**
LSMOP8	3	500	5.5092e-1 (6.99e-3) -	2.5499e-1 (4.40e-2) -	9.5821e-1 (3.20e-2) -	2.0243e-1 (9.58e-2) -	3.0093e-1 (5.65e-2) -	**8.2831e-2 (7.77e-4)**
		1000	5.3385e-1 (6.78e-3) -	2.2844e-1 (3.50e-2) -	9.4605e-1 (3.79e-2) -	2.3220e-1 (1.88e-1) -	2.9133e-1 (9.79e-2) -	**7.4195e-2 (4.37e-4)**
		2000	5.2963e-1 (6.42e-3) -	2.1269e-1 (1.60e-2) -	9.5304e-1 (4.07e-3) -	2.7366e-1 (2.47e-1) -	4.3041e-1 (1.60e-1) -	**7.0155e-2 (3.10e-4)**
		5000	5.2427e-1 (6.07e-3) -	2.1161e-1 (1.39e-2) -	9.4425e-1 (2.46e-2) -	4.7586e-1 (3.02e-1) -	5.4896e-1 (4.05e-2) -	**6.8185e-2 (3.35e-4)**
LSMOP9	3	500	8.9703e-1 (1.06e-1) -	**5.8500e-1 (1.72e-3)** +	3.1590e+0 (5.10e-1) -	1.5842e+1 (5.05e+0) -	1.8437e+0 (1.87e-1) -	5.8739e-1 (4.22e-3)
		1000	8.9095e-1 (9.12e-2) -	**5.8149e-1 (2.40e-3)** +	1.0155e+1 (1.51e+0) -	2.8078e+1 (7.74e+0) -	3.9049e+0 (1.88e+0) -	5.8744e-1 (3.60e-3)
		2000	1.8826e+0 (7.61e-1) -	**5.8062e-1 (1.81e-3)** +	2.7680e+1 (2.19e+0) -	4.6896e+1 (6.72e+0) -	9.7441e+0 (5.38e+0) -	5.8688e-1 (3.65e-3)
		5000	1.3011e+1 (3.61e+0) -	**5.7979e-1 (2.16e-3)** +	5.5360e+1 (1.88e+0) -	5.6444e+1 (1.11e+1) -	1.8209e+1 (6.22e+0) -	5.8520e-1 (4.56e-3)
+/-/=			0/36/0	6/29/1	0/36/0	0/33/3	0/36/0	

Table 4
HV results obtained by TSEAODS and five comparison algorithms on three objectives with different dimensions.

Problem	M	D	LMOCSO	LMOEADS	FDV	DGEA	LERD	TSEAODS
LSMOP1	3	500	0.000e+0 (0.00e+0) -	2.3272e-1 (4.87e-2) -	0.000e+0 (0.00e+0) -	1.5988e-1 (9.21e-2) -	1.3550e-2 (3.71e-2) -	**3.9240e-1 (2.60e-2)**
		1000	0.000e+0 (0.00e+0) -	1.8805e-1 (5.28e-2) -	0.000e+0 (0.00e+0) -	1.0431e-1 (1.01e-1) -	1.9305e-3 (6.33e-3) -	**3.9203e-1 (2.91e-2)**
		2000	0.000e+0 (0.00e+0) -	1.7571e-1 (3.14e-2) -	0.000e+0 (0.00e+0) -	1.0347e-1 (7.88e-2) -	6.7658e-4 (2.12e-3) -	**3.7411e-1 (2.46e-2)**
		5000	0.000e+0 (0.00e+0) -	1.8966e-1 (4.29e-2) -	0.000e+0 (0.00e+0) -	1.0975e-1 (8.00e-2) -	0.000e+0 (0.00e+0) -	**3.8487e-1 (2.94e-2)**
LSMOP2	3	500	8.1894e-1 (5.49e-4) -	8.1444e-1 (4.21e-3) -	7.8622e-1 (3.00e-3) -	8.0488e-1 (5.21e-3) -	8.0729e-1 (1.49e-3) -	**8.3563e-1 (6.00e-4)**
		1000	8.3233e-1 (4.07e-4) -	8.2066e-1 (3.92e-3) -	8.0835e-1 (3.79e-3) -	8.1645e-1 (4.88e-3) -	8.2707e-1 (5.97e-4) -	**8.4143e-1 (3.16e-4)**
		2000	8.3953e-1 (2.62e-4) -	8.2483e-1 (3.75e-3) -	8.2170e-1 (2.09e-3) -	8.2459e-1 (3.07e-3) -	8.3726e-1 (1.92e-4) -	**8.4447e-1 (3.06e-4)**
		5000	8.4451e-1 (9.18e-5) -	8.2770e-1 (3.28e-3) -	8.2820e-1 (3.06e-3) -	8.2800e-1 (3.15e-3) -	8.4358e-1 (2.11e-4) -	**8.4657e-1 (1.43e-4)**
LSMOP3	3	500	0.000e+0 (0.00e+0) -	**9.0911e-2 (5.38e-4)** +	0.000e+0 (0.00e+0) -	8.1837e-2 (2.80e-2) =	0.000e+0 (0.00e+0) -	9.0400e-2 (1.48e-3)
		1000	0.000e+0 (0.00e+0) -	**9.0955e-2 (1.48e-4)** +	0.000e+0 (0.00e+0) -	7.2740e-2 (3.73e-2) =	0.000e+0 (0.00e+0) -	9.0736e-2 (6.50e-4)
		2000	0.000e+0 (0.00e+0) -	**9.0431e-2 (1.37e-3)** =	0.000e+0 (0.00e+0) -	6.8190e-2 (4.04e-2) -	0.000e+0 (0.00e+0) -	8.8771e-2 (9.09e-3)
		5000	0.000e+0 (0.00e+0) -	9.0431e-2 (1.51e-3) =	0.000e+0 (0.00e+0) -	4.9857e-2 (4.63e-2) -	0.000e+0 (0.00e+0) -	**9.0951e-2 (2.25e-5)**
LSMOP4	3	500	7.0019e-1 (2.11e-3) -	7.5330e-1 (4.26e-3) -	6.3146e-1 (5.27e-3) -	7.3669e-1 (4.31e-3) -	7.1085e-1 (5.85e-3) -	**7.8369e-1 (4.38e-3)**
		1000	7.6951e-1 (6.81e-4) -	7.9438e-1 (2.51e-3) -	7.2755e-1 (4.03e-3) -	7.8528e-1 (3.88e-3) -	7.6593e-1 (3.69e-3) -	**8.1450e-1 (1.87e-3)**
		2000	8.0873e-1 (2.83e-4) -	8.1247e-1 (3.85e-3) -	7.8038e-1 (2.71e-3) -	8.0787e-1 (4.40e-3) -	8.0520e-1 (1.60e-3) -	**8.2991e-1 (8.79e-4)**
		5000	8.3333e-1 (2.07e-4) -	8.2285e-1 (3.29e-3) -	8.1337e-1 (4.14e-3) -	8.2231e-1 (3.46e-3) -	8.3108e-1 (5.28e-4) -	**8.4029e-1 (4.57e-4)**
LSMOP5	3	500	0.000e+0 (0.00e+0) -	**3.4492e-1 (7.95e-4)** +	0.000e+0 (0.00e+0) -	2.8203e-1 (1.34e-1) -	2.5083e-2 (8.50e-2) -	3.3253e-1 (1.57e-3)
		1000	0.000e+0 (0.00e+0) -	**3.4511e-1 (5.22e-4)** +	0.000e+0 (0.00e+0) -	3.1189e-1 (8.56e-2) -	0.000e+0 (0.00e+0) -	3.3195e-1 (1.88e-3)
		2000	0.000e+0 (0.00e+0) -	**3.4479e-1 (6.91e-4)** +	0.000e+0 (0.00e+0) -	2.4490e-1 (1.41e-1) =	0.000e+0 (0.00e+0) -	3.3101e-1 (1.54e-3)
		5000	0.000e+0 (0.00e+0) -	**3.4474e-1 (3.65e-4)** +	0.000e+0 (0.00e+0) -	2.8242e-1 (1.25e-1) -	0.000e+0 (0.00e+0) -	3.3141e-1 (2.00e-3)
LSMOP6	3	500	0.000e+0 (0.00e+0) -	9.2995e-3 (5.02e-3) -	0.000e+0 (0.00e+0) -	2.2478e-2 (2.23e-2) -	0.000e+0 (0.00e+0) -	**4.5950e-2 (1.60e-2)**
		1000	0.000e+0 (0.00e+0) -	9.1247e-3 (4.30e-3) -	0.000e+0 (0.00e+0) -	1.4284e-2 (2.18e-2) -	0.000e+0 (0.00e+0) -	**3.9109e-2 (2.20e-2)**
		2000	0.000e+0 (0.00e+0) -	9.3610e-3 (3.57e-3) -	0.000e+0 (0.00e+0) -	8.3652e-3 (1.03e-2) -	0.000e+0 (0.00e+0) -	**5.7634e-2 (1.79e-2)**
		5000	0.000e+0 (0.00e+0) -	1.0741e-2 (5.22e-3) -	0.000e+0 (0.00e+0) -	7.7018e-3 (9.75e-3) -	0.000e+0 (0.00e+0) -	**6.1654e-2 (2.26e-2)**
LSMOP7	3	500	**5.3799e-2 (4.51e-2)** +	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	0.000e+0 (0.00e+0) =	1.8798e-3 (6.92e-3) =	0.000e+0 (0.00e+0)
		1000	**3.1860e-2 (3.62e-2)** +	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	6.7017e-3 (6.14e-3) -	4.9395e-4 (2.21e-3) -	1.6537e-2 (7.18e-3)
		2000	1.7739e-2 (1.91e-2) -	4.1599e-2 (3.40e-3) -	0.000e+0 (0.00e+0) -	4.1638e-2 (9.90e-3) -	2.0305e-2 (8.52e-3) -	**4.8031e-2 (1.64e-2)**
		5000	5.3255e-2 (4.46e-3) -	7.1890e-2 (1.00e-3) -	4.2711e-2 (6.10e-4) -	7.1337e-2 (3.72e-3) -	6.2033e-2 (2.79e-3) -	**7.5522e-2 (3.63e-5)**
LSMOP8	3	500	6.8927e-2 (4.57e-3) -	3.8834e-1 (8.43e-3) -	5.2902e-2 (1.11e-3) -	3.9423e-1 (7.78e-2) -	3.5592e-1 (1.64e-2) -	**4.7693e-1 (9.78e-4)**
		1000	8.4564e-2 (5.14e-3) -	3.9468e-1 (6.36e-3) -	6.7880e-2 (7.77e-4) -	3.7222e-1 (1.28e-1) -	3.0675e-1 (1.27e-1) -	**4.9008e-1 (7.18e-4)**
		2000	9.0094e-2 (4.23e-3) -	3.9389e-1 (5.23e-3) -	7.5833e-2 (4.54e-4) -	3.4638e-1 (1.55e-1) -	1.7249e-1 (1.26e-1) -	**4.9639e-1 (4.77e-4)**
		5000	9.5068e-2 (4.14e-3) -	3.9588e-1 (3.85e-3) -	8.1133e-2 (2.16e-4) -	2.4030e-1 (1.72e-1) -	9.5702e-2 (7.67e-3) -	**4.9972e-1 (6.31e-4)**
LSMOP9	3	500	5.2647e-2 (5.14e-3) -	**1.9057e-1 (3.42e-4)** +	9.3050e-5 (4.16e-4) -	0.000e+0 (0.00e+0) -	3.7314e-2 (2.14e-2) -	1.8970e-1 (1.37e-3)
		1000	4.9386e-2 (9.37e-3) -	**1.9086e-1 (4.02e-4)** +	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	3.3315e-3 (1.49e-2) -	1.9015e-1 (1.15e-3)
		2000	1.4091e-2 (1.32e-2) -	**1.9095e-1 (3.02e-4)** +	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	1.9028e-1 (8.68e-4)
		5000	0.000e+0 (0.00e+0) -	**1.9111e-1 (3.42e-4)** +	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	0.000e+0 (0.00e+0) -	1.9049e-1 (9.41e-4)
+/-/=			2/34/0	10/23/3	0/35/1	0/32/4	0/35/1	

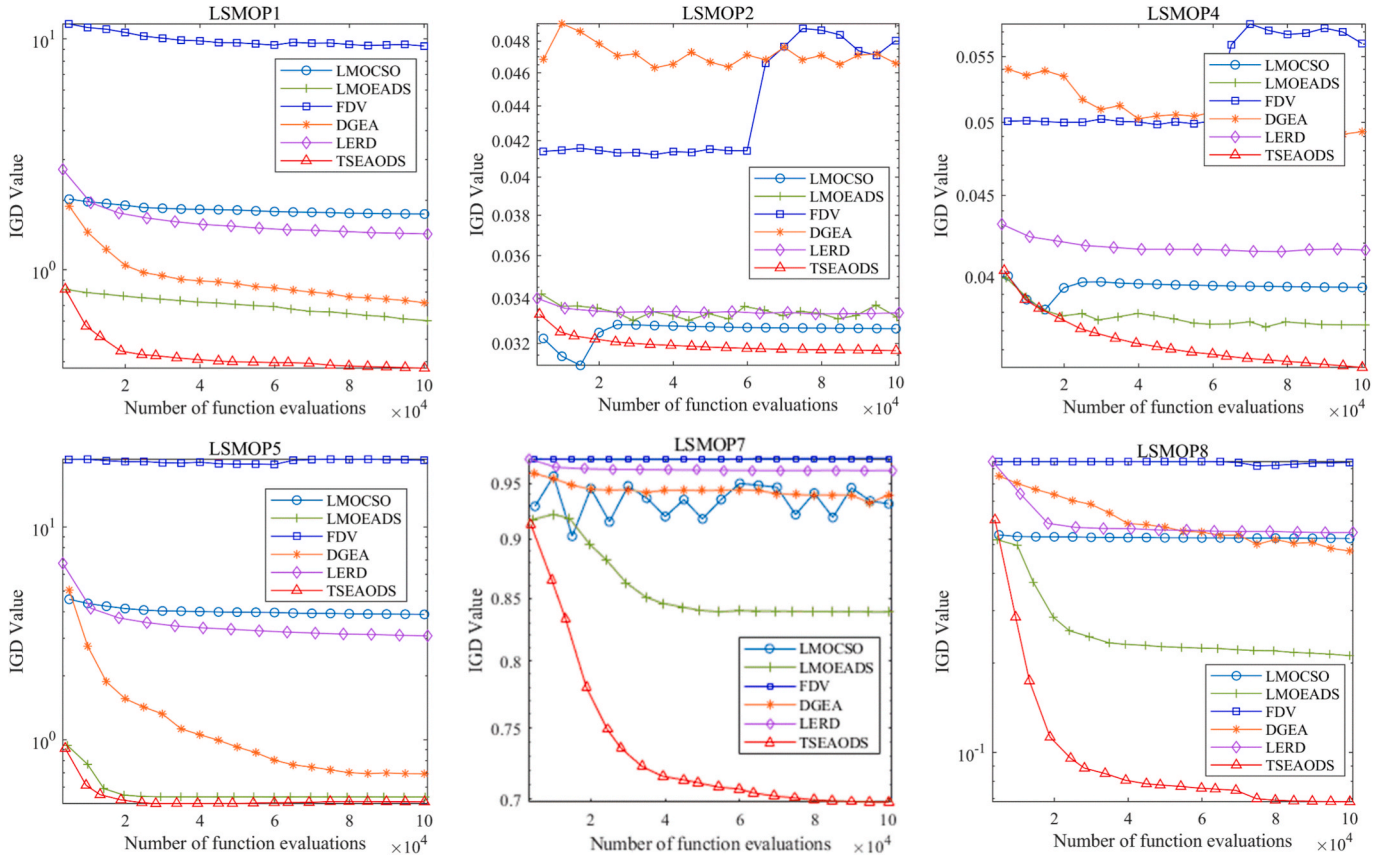


Fig. 7. Curves of the mean IGD value obtained by LMOCSO, LMOEADS, FDV, DGEA, LERD and TSEAODS on three-objective LSMOP1, LSMOP2, LSMOP4, LSMOP5, LSMOP7 and LSMOP8 with 5000 decision variables.

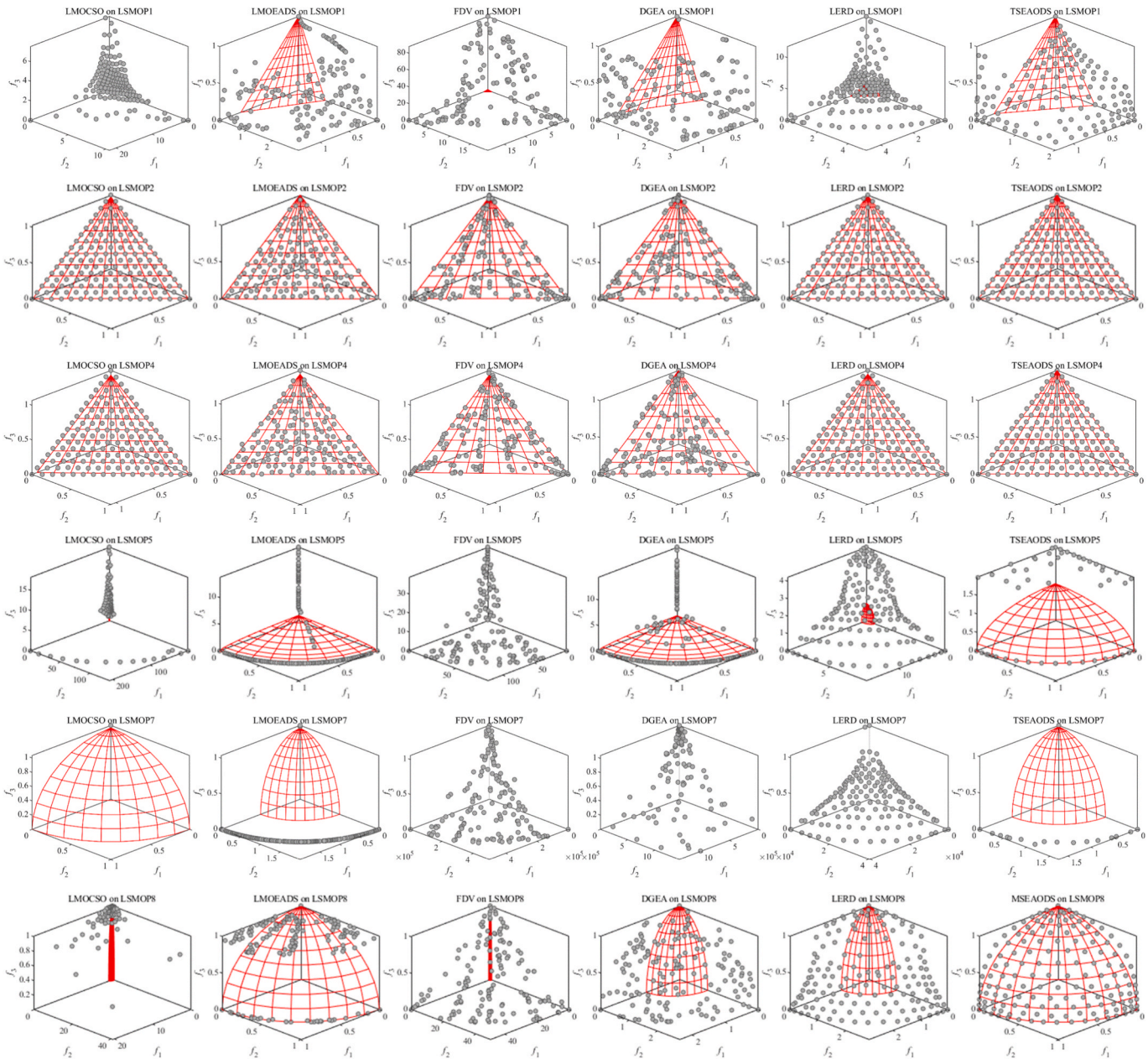


Fig. 8. The final solution sets obtained by LMOCSO, LMOEADS, FDV, DGEA, LERD and TSEAODS on three-objective LSMOP1, LSMOP2, LSMOP4, LSMOP5, LSMOP7 and LSMOP8 with 5000 decision variables.

strategy, performs better than TSEAODS in IGD values on only 6 out of 36 test problems and in HV values on 10 problems. In brief, it demonstrates the effectiveness of our proposed two-stage algorithm framework, offset direction sampling strategy, and dual-layer competition strategy.

Fig. 7 illustrates the average IGD value curves for each algorithm. It is evident that TSEAODS converges the fastest among the compared algorithms, achieving a lower IGD value early in the evolution and sustaining this level of convergence due to its two-stage algorithm framework. TSEAODS demonstrates similar performance to other algorithms on linear problems LSMOP1, LSMOP2, and LSMOP4, as observed in two objectives scenarios, whereas it exhibits significant improvements and advantages in addressing unimodal problem LSMOP5, multimodal problem LSMOP7, and hybrid problem LSMOP8.

To further illustrate the performance of TSEAODS relative to other algorithms on three-objective LSMOP problems, Fig. 8 presents a comparison of the results of algorithms with the true PF for problems LSMOP1, LSMOP2, LSMOP4, LSMOP5, LSMOP7, and LSMOP8 in the context of three-objective 5000-dimensional test problems. Fig. 8 indicates that TSEAODS exhibits a uniform population distribution and converges well to the true PF on LSMOP2 and LSMOP4. In contrast, the results from LMOEADS, FDV, and DGEA are not uniformly distributed on the true PF, whereas LMOCSO and LERD have relatively uniform distributions, but their convergence is not as effective as that of TSEAODS. For LSMOP1, TSEAODS is less close to the true PF, although there is an improvement in population distribution uniformity compared to other algorithms. For LSMOP5, the proposed algorithm identifies only a few solutions that converge to the true PF. On LSMOP7, none of the algorithms successfully converge to the true PF, suggesting that existing algorithms perform inadequately on multimodal problems. For LSMOP8, TSEAODS achieves a better-distributed population compared to other algorithms, though some solutions do not completely converge to the true PF.

3) Effectiveness Analysis of offset direction sampling strategy. To

Table 5
The IGD comparison results between TSEAODS and TSEADS.

Problem	M	D	TSEADS	TSEAODS
LSMOP1	3	500	5.8530e-1 (4.85e-2) -	**3.6293e-1 (2.88e-2)**
		1000	6.3493e-1 (3.82e-2) -	**3.6489e-1 (3.71e-2)**
		2000	6.6367e-1 (4.62e-2) -	**3.6359e-1 (3.11e-2)**
		5000	6.7274e-1 (4.02e-2) -	**3.6662e-1 (3.15e-2)**
LSMOP2	3	500	4.4950e-2 (5.63e-4) -	**3.8316e-2 (5.70e-4)**
		1000	3.7669e-2 (3.73e-4) -	**3.4303e-2 (2.57e-4)**
		2000	3.4215e-2 (1.09e-4) -	**3.2615e-2 (1.28e-4)**
		5000	3.2027e-2 (4.41e-5) -	**3.1671e-2 (5.36e-5)**
LSMOP3	3	500	8.6254e-1 (9.01e-3) =	**8.6084e-1 (5.31e-4)**
		1000	8.6084e-1 (1.66e-3) -	**8.6072e-1 (2.64e-4)**
		2000	8.6076e-1 (7.54e-4) =	8.6062e-1 (4.98e-5)
		5000	8.6299e-1 (9.26e-3) =	**8.6090e-1 (9.76e-4)**
LSMOP4	3	500	1.0777e-1 (2.10e-3) -	**8.7194e-2 (2.81e-3)**
		1000	6.6425e-2 (1.04e-3) -	**5.8156e-2 (1.51e-3)**
		2000	4.6276e-2 (5.05e-4) -	**4.2889e-2 (5.80e-4)**
		5000	3.5873e-2 (2.75e-4) -	**3.4989e-2 (2.92e-4)**
LSMOP5	3	500	5.1782e-1 (2.15e-2) -	**4.6662e-1 (1.02e-2)**
		1000	5.3748e-1 (6.90e-3) -	**4.8977e-1 (6.31e-3)**
		2000	5.3974e-1 (4.48e-3) -	**5.0355e-1 (3.40e-3)**
		5000	5.4071e-1 (2.61e-3) -	**5.1513e-1 (1.29e-3)**
LSMOP6	3	500	7.1074e-1 (1.61e-2) -	**6.5930e-1 (4.53e-2)**
		1000	7.1904e-1 (1.69e-2) -	**6.6716e-1 (7.23e-2)**
		2000	7.3209e-1 (1.12e-1) -	**6.7115e-1 (6.70e-2)**
		5000	7.1571e-1 (1.55e-2) =	**6.9238e-1 (7.18e-2)**
LSMOP7	3	500	8.9648e-1 (7.87e-2) -	**7.9611e-1 (1.82e-2)**
		1000	1.0283e+0 (2.27e-1) -	**7.7988e-1 (1.24e-1)**
		2000	1.1338e+0 (2.40e-1) -	**7.2421e-1 (6.82e-2)**
		5000	1.2634e+0 (1.69e-1) -	**6.9807e-1 (4.06e-3)**
LSMOP8	3	500	2.7020e-1 (4.59e-2) -	**8.2831e-2 (7.77e-4)**
		1000	2.2736e-1 (5.57e-2) -	**7.4195e-2 (4.37e-4)**
		2000	2.1286e-1 (2.88e-2) -	**7.0155e-2 (3.10e-4)**
		5000	2.2465e-1 (1.58e-2) -	**6.8185e-2 (3.35e-4)**
LSMOP9	3	500	5.8899e-1 (2.48e-3) =	5.8739e-1 (4.22e-3)
		1000	5.8725e-1 (3.09e-3) =	5.8744e-1 (3.60e-3)
		2000	5.8810e-1 (3.18e-3) =	**5.8688e-1 (3.65e-3)**
		5000	5.8657e-1 (3.87e-3) =	5.8520e-1 (4.56e-3)
+/-/=			0/28/8	

validate the effectiveness of the offset direction sampling strategy, a comparative experiment was conducted between TSEAODS and its variant, TSEADS, on LSMOP1-9 test problems with 3 objectives and decision variable dimensions of 500, 1000, 2000, and 5000. TSEADS represents the variant where no offset is applied to the search directions, and each high-quality solution generates only two search directions. Table 5 summarizes the average IGD values obtained by both algorithms in the experiments. The results show that TSEAODS consistently outperforms TSEADS across all test problems. Introducing offset to the search directions improves the likelihood of their intersection with PS, thereby enhancing the quality of sampled solutions and the convergence of the population. These findings demonstrate that the proposed offset direction sampling strategy is effective and significantly improves the performance and efficiency of the algorithm.

4) Effectiveness Analysis of dual-layer competition strategy. To validate the effectiveness of the dual-layer competition strategy, we designed a variant algorithm, SSEAODS, which removes the precise convergence stage of TSEAODS and retains only the offset direction sampling strategy. Comparative experiments were conducted using TSEAODS and SSEAODS on LSMOP1-9 test problems with three objectives and decision variable dimensions of 500, 1000, 2000, and 5000. Table 6 summarizes the average IGD values obtained by both algorithms. The results show that TSEAODS achieves better IGD performance than SSEAODS on half of the test problems, highlighting the importance of the dual-layer competition strategy in enhancing population convergence.

The dual-layer competition strategy applies non-dominated sorting to the population, where individuals in higher non-dominated ranks typically have lower quality. The population is divided into elite and non-elite layers. Individuals from lower non-dominated ranks are added to the elite layer until it comprises half the population, while the remaining individuals are assigned to the non-elite layer. Two superior individuals are identified through competition between the layers to guide the evolution of the non-elite layer and maintain population convergence. Subsequently, polynomial mutation is applied to the entire population to prevent premature convergence. The experimental results demonstrate that the dual-layer competition strategy makes a significant contribution to the overall performance improvement of the algorithm.

5) Parameter sensitivity analysis. To investigate the impact of the phase proportion parameter Rate on algorithmic performance, we conducted a sensitivity analysis by varying its values. The parameter Rate controls the allocation of function evaluations between the two stages: the rapid convergence stage and the precise convergence stage. Experiments were conducted on three-objective LSMOP1-9 test instances with 2000 decision variables, with Rate set at 0.3, 0.5, and 0.7. As shown in

Table 6
The IGD comparison results between TSEAODS and SSEAODS.

Problem	M	D	SSEAODS	TSEAODS
LSMOP1	3	500	3.7784e-1 (2.66e-2) -	**3.6293e-1 (2.88e-2)**
		1000	3.7504e-1 (3.05e-2) -	**3.6489e-1 (3.71e-2)**
		2000	3.9259e-1 (2.96e-2) -	**3.6359e-1 (3.11e-2)**
		5000	3.8559e-1 (3.28e-2) -	**3.6662e-1 (3.15e-2)**
LSMOP2	3	500	3.8567e-2 (3.89e-4) -	**3.8316e-2 (5.70e-4)**
		1000	3.4454e-2 (2.74e-4) -	**3.4303e-2 (2.57e-4)**
		2000	3.2725e-2 (1.89e-4) -	**3.2615e-2 (1.28e-4)**
		5000	**3.1622e-2 (4.75e-5)** +	3.1671e-2 (5.36e-5)
LSMOP3	3	500	8.6112e-1 (2.31e-3) =	8.6084e-1 (5.31e-4)
		1000	8.6076e-1 (4.94e-4) =	8.6072e-1 (2.64e-4)
		2000	8.6068e-1 (3.76e-4) =	8.6062e-1 (4.98e-5)
		5000	8.6063e-1 (1.38e-4) =	8.6090e-1 (9.76e-4)
LSMOP4	3	500	9.0602e-2 (2.78e-3) -	**8.7194e-2 (2.81e-3)**
		1000	5.9606e-2 (1.72e-3) -	**5.8156e-2 (1.51e-3)**
		2000	4.3799e-2 (1.09e-3) -	**4.2889e-2 (5.80e-4)**
		5000	3.5058e-2 (3.00e-4) =	3.4989e-2 (2.92e-4)
LSMOP5	3	500	4.6590e-1 (1.09e-2) =	4.6662e-1 (1.02e-2)
		1000	4.8849e-1 (6.33e-3) =	4.8977e-1 (6.31e-3)
		2000	**4.9843e-1 (7.89e-3)** +	5.0355e-1 (3.40e-3)
		5000	**5.1156e-1 (5.47e-3)** +	5.1513e-1 (1.29e-3)
LSMOP6	3	500	**6.2230e-1 (6.21e-2)** +	6.5930e-1 (4.53e-2)
		1000	**5.9971e-1 (2.18e-2)** +	6.6716e-1 (7.23e-2)
		2000	**6.1239e-1 (2.98e-2)** +	6.7115e-1 (6.70e-2)
		5000	**6.0408e-1 (3.66e-2)** +	6.9238e-1 (7.18e-2)
LSMOP7	3	500	8.0302e-1 (7.32e-3) -	**7.9611e-1 (1.82e-2)**
		1000	7.3582e-1 (4.05e-3) =	7.7988e-1 (1.24e-1)
		2000	**7.0943e-1 (8.73e-3)** +	7.2421e-1 (6.82e-2)
		5000	**6.9534e-1 (3.84e-3)** +	6.9807e-1 (4.06e-3)
LSMOP8	3	500	8.8570e-2 (3.40e-3) -	**8.2831e-2 (7.77e-4)**
		1000	7.8907e-2 (2.27e-3) -	**7.4195e-2 (4.37e-4)**
		2000	7.4957e-2 (2.80e-3) -	**7.0155e-2 (3.10e-4)**
		5000	7.2024e-2 (1.06e-3) -	**6.8185e-2 (3.35e-4)**
LSMOP9	3	500	5.8677e-1 (3.69e-3) =	5.8739e-1 (4.22e-3)
		1000	5.8782e-1 (3.43e-3) =	5.8744e-1 (3.60e-3)
		2000	5.8626e-1 (4.38e-3) =	5.8688e-1 (3.65e-3)
		5000	5.8620e-1 (3.48e-3) =	5.8520e-1 (4.56e-3)
+/-/=			9/15/12	

Table 7
The IGD results of TSEAODS under different Rate settings on three-objective problems with 2000 dimensions.

Problem	Rate=0.3	Rate=0.5	Rate=0.7
LSMOP1	**3.6055e-1 (3.59e-2)** +	3.7547e-1 (3.46e-2) =	3.6359e-1 (3.13e-2)
LSMOP2	3.2743e-2 (1.81e-4) -	3.2636e-2 (2.11e-4) =	**3.2615e-2 (1.28e-4)**
LSMOP3	8.6202e-1 (7.24e-3) =	8.6092e-1 (1.36e-3) =	**8.6062e-1 (4.98e-5)**
LSMOP4	4.3045e-2 (9.80e-4) =	4.2890e-2 (7.20e-4) =	4.2889e-2 (5.80e-4)
LSMOP5	5.0988e-1 (2.74e-3) -	5.0647e-1 (3.41e-3) -	**5.0355e-1 (3.40e-3)**
LSMOP6	7.1830e-1 (6.86e-2) -	6.9679e-1 (6.79e-2) =	**6.7115e-1 (6.70e-2)**
LSMOP7	8.0411e-1 (1.73e-1) -	7.5952e-1 (1.28e-1) =	**7.2421e-1 (6.82e-2)**
LSMOP8	**6.9955e-2 (4.72e-4)** +	6.9984e-2 (4.58e-4) +	7.0155e-2 (3.10e-4)
LSMOP9	5.8865e-1 (4.35e-3) =	5.8776e-1 (3.15e-3) =	**5.8688e-1 (3.65e-3)**
+/-/=	2/4/3	1/1/7	

Table 7, TSEAODS exhibits strong robustness to Rate, yet achieves optimal performance when Rate = 0.7. Furthermore, Fig. 9 illustrates the average computational time over 30 independent runs, indicating that Rate = 0.7 yields the lowest computational overhead. Considering both solution quality and efficiency, Rate = 0.7 is empirically justified. Notably, extreme configurations (e.g., Rate = 0.1 or 0.9) were excluded to prevent insufficient evaluations in either stage, which could compromise the effectiveness of the corresponding strategies.

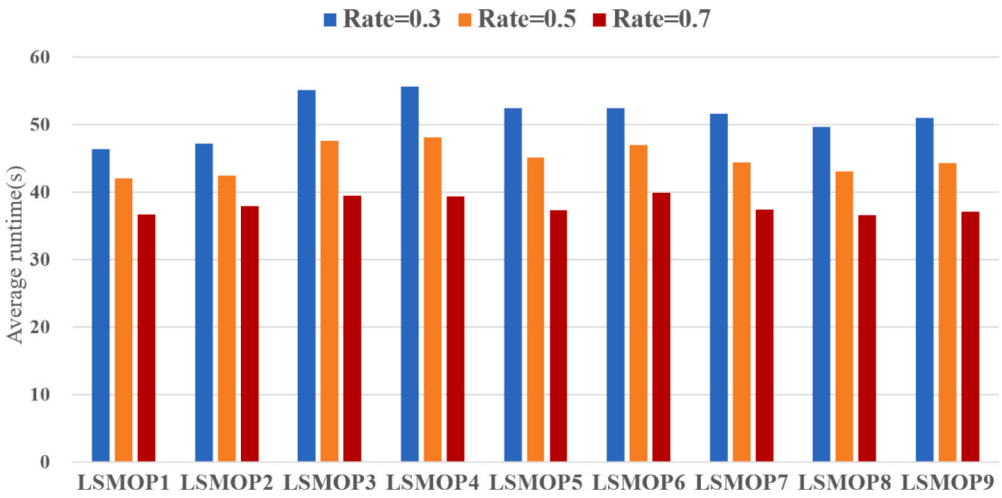


Fig. 9. Average runtime of TSEAODS over 30 runs under different Rate settings on three-objective problems with 2000 dimensions.

5. Conclusion

This paper proposes a two-stage large-scale multiobjective evolutionary algorithm based on offset direction sampling and dual-layer competition. The two-stage algorithm framework balances the search efficiency and quality of the solutions. The offset direction sampling strategy diversifies the original sampling directions to increase direction diversity; thus, the generated sampling solutions are more diverse. This enhances the likelihood of intersecting the PS plane, thereby guiding the population to converge rapidly. The dual-layer competition strategy compensates for the possible suboptimal solutions in the previous stage. All individuals in the non-elite layer learn from the two winning individuals, thereby improving the quality of the final solutions. Experimental results demonstrate that TSEAODS is competitive in overall performance compared to other algorithms.

Our future work focuses on enhancing TSEAODS to tackle large-scale multimodal problems, particularly neural network optimization. We plan to design an adaptive direction sampling method that dynamically adjusts the sampling directions based on problem characteristics and the current search state, aiming to endow systems with adaptability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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